

X-ray photoelectron spectroscopy (XPS) of Quinacridone (QA) on the Ag(100) surface

Focusing Laboratory Course

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1 Introduction

The interface of organic semiconductors with metal surfaces is of particular importance within the domain of organic electronics. The aforementioned organic electronics include, for example, organic light-emitting diodes (OLEDs), organic photovoltaics (OPVs) and organic field-effect transistors (OFETs).^[1]

One organic molecule that has been extensively studied in the context of organic electronics is quinacridone (QA). QA is an organic, prochiral molecule with a large π -conjugated system, which is known for its use as a dye in printer toners and varnishes.^[2] However, its potential extends beyond the domain of dyes, as previous studies demonstrated the application of QA in organic electronics.^[3–5] Specific applications for QA have been identified, including its use in OLEDs,^[6–8] OPVs^[9,10] and OFETs.^[11–14] In addition, QA has been shown to form well-defined structures on metal surfaces, such as the Ag(100), the Ag(111) and the Cu(111) surface.^[15–19]

A common structural motif of QA monolayers on surfaces^[16,19–21] and in its bulk structure^[22] are 1D molecular chains. The formation of the 1D molecular chains is driven by intermolecular hydrogen bonds.^[15,16]

In particular, QA on the Ag(100) surface forms two different phases, the α -phase and the β -phase. The α -phase consists of the previously described homochiral, parallel lying 1D molecular chains. This leads to a point-on-line (POL) structure for the α -phase. In contrast, the β -phase is characterized by a commensurate phase, which is thermodynamically more stable than the α -phase and forms irreversibly when the sample is heated. The β -phase has fewer hydrogen bonds and stronger adsorbate-substrate interactions. In addition to this, the β -phase has proposed to be heterochiral, necessitating a change of the handedness of the molecules during the phase transition.^[15,16]

The phase transition and the differences between the two phases are of particular interest, as the change in the binding situation of the different atoms are not yet known.

The present study is focused on how the binding situation of QA on the Ag(100) surface is impacted by the interplay between intermolecular interactions via hydrogen bonds and adsorbate-substrate interactions across the two distinct phases. Specifically, the quantity of the hydrogen bonds and the specific atoms that are involved in the adsorbate-surface interactions are of interest. Furthermore, the diversity of chemical

environments that exist for the different atoms in the molecule and the subsequent alterations that occur upon phase transition, are of particular relevance.

For this purpose, XPS is used as the primary technique. XPS provides insights into the binding energies of the core level electrons for the different types of atoms present in the adsorbate. Consequently, the binding energies of the core level electrons give information about the binding situation of the different substrate atoms on the Ag(100) surface. However, the details about the binding situation of the adsorbate are not accessible with the previously used techniques, such as scanning tunneling microscopy (STM) and low energy electron diffraction (LEED), as they only provide information about the molecular arrangement.

The XPS spectra offer new insights regarding the priorly mentioned binding situation. Previous STM and LEED studies proposed 2 hydrogen bonds per molecule in the α -phase and 1.5 hydrogen bonds per molecule in the β -phase.^[15,16] Similarly, the present study determined 2 hydrogen bonds per molecule in the α -phase, but in contrast to previous studies, only 1 hydrogen bond per molecule in the β -phase. This finding necessitates a new structure model, a subject that will be examined in the subsequent chapters.

2 Literature

2.1 Quinacridone (QA)

The examined adsorbate quinacridone (QA) is a high symmetric, prochiral organic molecule with the IUPAC name 5,12-dihydro-quin[2,3-*b*]acridine-7,14-dione and the sum formula $C_{20}H_{12}N_2O_2$. Organic crystals exhibit a color which can range from red to violet.^[23,24]

QA is most prominently recognized as the parent structure of a group of pigments, most notably Pigment Violet 19 (C.I. 73900). This particular pigment is extensively utilized in industrial applications due to its remarkable color properties and stability.^[23,24] The molecular structure of QA is shown in Figure 2.1.

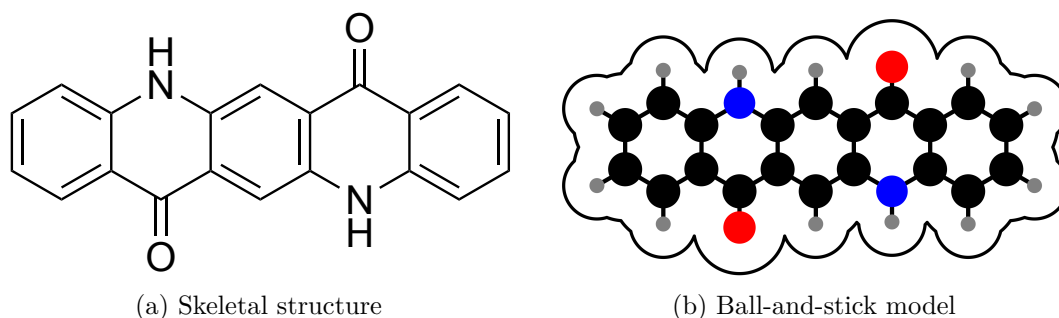


Figure 2.1: Molecular structure of quinacridone (QA).

The molecule is classified as a heterocyclic compound. Its structure is characterized by a conjugated ring system that induces significant rigidity and planarity to the molecule.^[25] QA is of the symmetry group C_{2h} and has a molar mass of 312.32 g/mol. The molecule is a crystalline solid with four known different crystal structures under standard conditions.^[22,26]

2.2 Structures of Quinacridone (QA) on the Ag(100) surface

For QA deposited onto the Ag(100) surface, two structural phases were observed to form: the α - and the β -phase. A thorough examination of these two phases has been conducted by N. Humberg.^[15,16]

The formation of the α -phase occurs when the QA molecules are deposited onto the Ag(100) surface at a sample temperature of 300 K. This phase is composed of parallel molecular chains that are flat-lying with the π -system being oriented parallel to the Ag(100) surface. The QA molecules are connected via intermolecular hydrogen bonds between the keto groups of one molecule and the amine groups of another molecule, resulting in a total of 4 hydrogen bonds. It was determined that the domains under consideration are homochiral, a finding that suggests a high degree of mobility for the QA molecules on Ag(100) surface at 300 K.

In addition, the superstructure matrix of the α -phase for a complete monolayer (ML) was determined using a spot profile analysis low energy electron diffraction (SPA-LEED) pattern and STM results. The following superstructure matrix was found

$$\mathbf{M}_\alpha = \begin{pmatrix} 2 & 1.25 \\ -3 & 4.80 \end{pmatrix},$$

which shows that the structure is of the POL type. This means that every lattice point of the superstructure falls on a substrate lattice line of the $[10]$ direction of the Ag(100) surface. It should be noted that the intermolecular distance b_1 between the molecules of a chain are determined by the intermolecular hydrogen bonds and does not change with the coverage. The distance between two molecular chains b_2 increases with decreasing coverage.

The STM images, the SPA-LEED pattern and the current structure model of QA on the Ag(100) surface in the α -phase is illustrated in Figure 2.2.

The phase transition from the α -phase to the β -phase occurs when the sample is heated up to 500 K for 15 minutes. The phase transition is irreversible. This means that the α -phase is metastable and is stabilized by a kinetic barrier. This kinetic barrier corresponds to the energy required to break the hydrogen bonds between the QA molecules in the α -phase.

For the β -phase, the investigation revealed a 2D network of molecular chains, which are comprised of dimers and manifest periodic indents. The current structure model proposes a single hydrogen bond between the dimers instead of theoretical two possible hydrogen bonds. The two molecules of the dimer form two hydrogen bonds with each other, resulting in 1.5 hydrogen bonds per molecule.

Once more, employing a SPA-LEED pattern and STM results, the superstructure matrix of the β -phase for a complete ML was determined as

$$\mathbf{M}_\beta = \begin{pmatrix} 4 & 3 \\ -5 & 3 \end{pmatrix},$$

which means that the structure is commensurate. In comparison to the POL type of the α -phase, the commensurate structure of the β -phase indicates that the superstructure for the β -phase is aligned with the substrate in a way that the periodicity

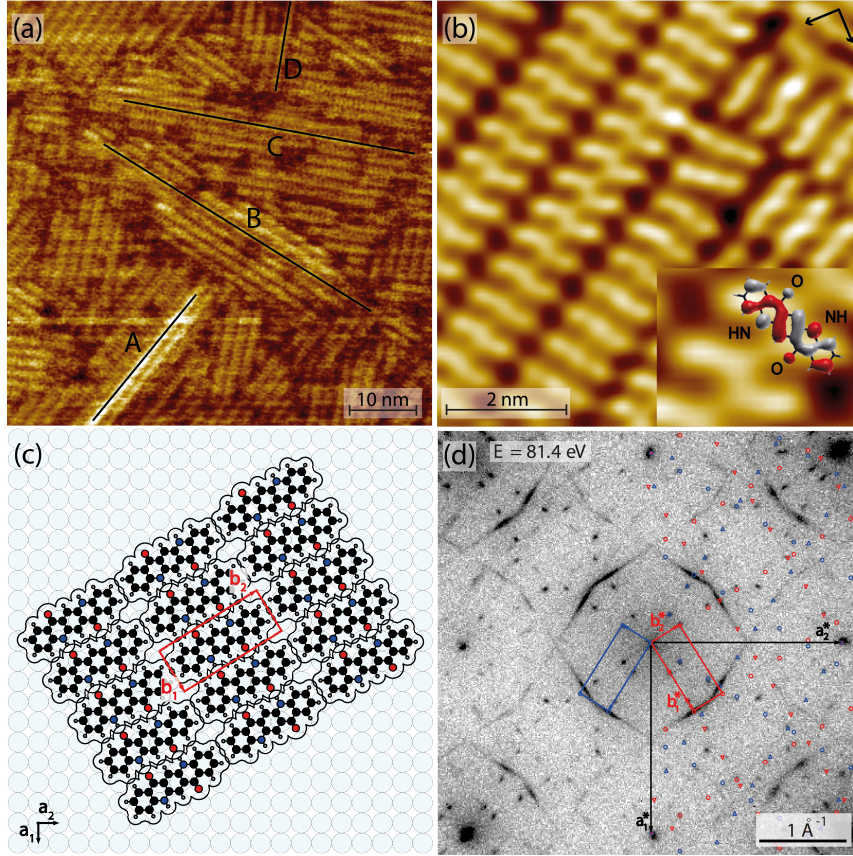


Figure 2.2: Current structure of complete ML of QA on Ag(100) in the α -phase. The image shows an overview STM image (a), a close-up STM image of the molecular arrangement (b), the SPA-LEED pattern (c) and the structure model (d). The image is taken from reference [16].

of the superstructure is a multiple of the substrate lattice constant. In general, the commensurate structure of the β -phase indicates a stronger interaction with the substrate and provides energy gain compared to the α -phase.^[27]

The STM images, the SPA-LEED pattern and the current structure model of QA on the Ag(100) surface in the β -phase is illustrated in Figure 2.3.

The current structure model proposed by N. Humberg^[15,16] already allow for the formulation of a prediction of the XPS spectra. This prediction is based on the different molecular arrangements and interactions in the α -phase and β -phase. In sum, the adsorbate-substrate interaction in the β -phase should be stronger than in the α -phase, leading to lower binding energies (BEs) for all components in the β -phase.

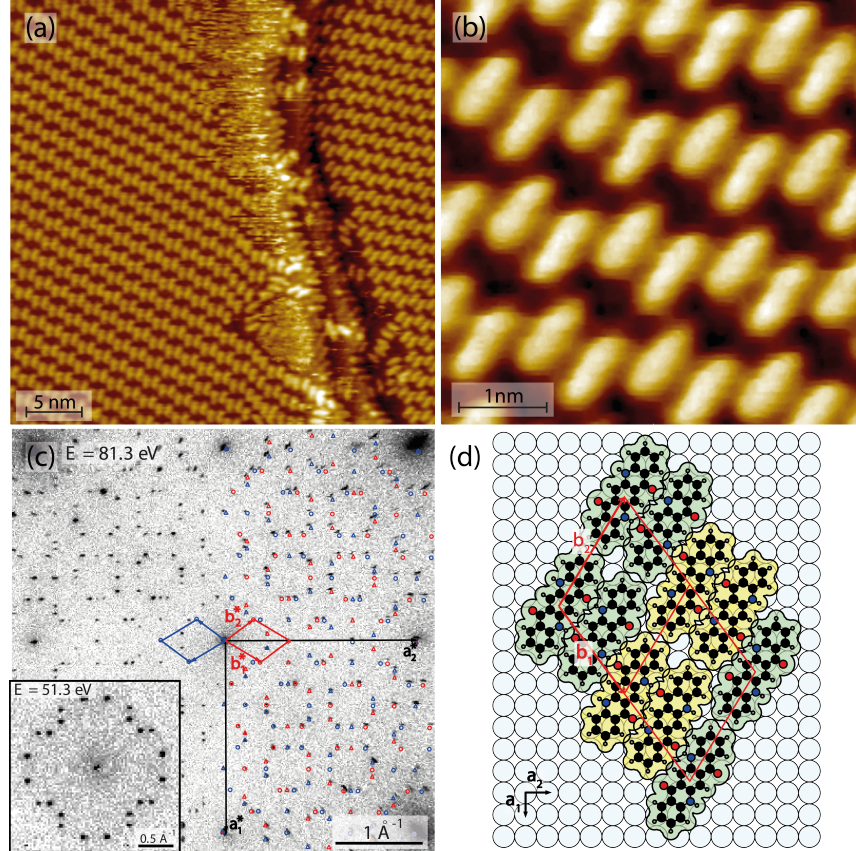


Figure 2.3: Current structure of complete ML of QA on Ag(100) in the β -phase. The image shows an overview STM image (a), a close-up STM image of the molecular arrangement (b), the SPA-LEED image (c) and the structure model (d). The image is taken from reference [16].

Furthermore, $3/4$ oxygen and nitrogen atoms in the β -phase are involved in hydrogen bonds, while in the α -phase all oxygen and nitrogen atoms are involved in hydrogen bonds. This should result in one peak for the oxygen and nitrogen atoms in the α -phase and two distinct peaks with an area ratio of $3 : 1$ for the oxygen and nitrogen atoms in the β -phase.

3 Theory

3.1 X-ray photoelectron spectroscopy (XPS)

X-ray photoelectron spectroscopy (XPS) utilizes the ejection of electrons from bound states to the continuum by photon excitation, called the photoelectric effect.^[28,29] The photon energies ($E_\gamma = h\nu$) of x-rays are similar to the binding energy (E_B) of deep core electrons and are therefore used for the excitation. The experiments for this work were performed with soft x-rays with an energy between 100 - 2100 eV. For each core level, E_γ is chosen to be slightly higher than E_B , resulting in the emission of photoelectrons with a kinetic energy (E_{kin}) of approximately 100 eV. This leads to a high surface sensitivity of the method, as the mean free path of these electrons is only a few nanometers.^[30]

The photoemission process is illustrated in Figure 3.1. The x-ray photons are absorbed by an electron in the core levels of the atom, which results in the ejection of a photoelectron. The energy of the emitted photoelectron is determined by the difference between the photon energy, which is tunable, and the binding energy of the electron in its initial state. Subsequently, the energy of the emitted photoelectron is detected and analyzed to provide information about the change of energy due to the bonding environment and its oxidation state. This phenomenon is referred to as chemical shift.^[31]

The kinetic energy of the photoelectron measured by the analyzer is relative to the Fermi level E_F and given by:

$$E'_{\text{kin}} = E_{\text{kin}} + \phi_{\text{sample}}, \quad (3.1)$$

where E_{kin} is the kinetic energy of the photoelectron and ϕ_{sample} is the work function of the sample. The kinetic energy of the photoelectron is defined as:

$$E_{\text{kin}} = E_\gamma - E_B - \phi_{\text{sample}}, \quad (3.2)$$

where $E_\gamma = h\nu$ is the photon energy and E_B is the binding energy of the electron.

E_B of an electron is defined as the difference in energies between the atom with n electrons and the ion with $n - 1$ electrons:

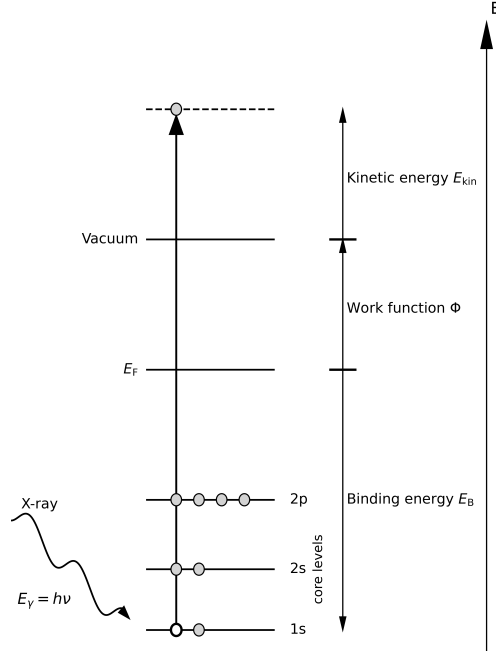


Figure 3.1: Schematic drawing of the photoemission process of an electron in the 1s orbital upon x-ray excitation. The image was adapted using reference [32].

$$E_B = E_f(n-1) - E_i(n), \quad (3.3)$$

As seen in Equation 3.3, E_B of photoelectrons depends on initial and final state effects.

The initial state effects manifest for the neutral unexcited atom, due to its interaction with its immediate electronic environment, also referred to as the chemical environment. In addition, final state effects emerge from the correlation between the photoemission process and the ionized final state. The rapid nature of the photoemission process may preclude the system from attaining adiabatic equilibrium.^[33]

The final state effects are characterized by the presence of satellites with reduced kinetic energy, correlating to increased E_B . This is exemplified by phenomena such as shake-up lines or asymmetric line shapes of the signal, which are indicative of photoemission from the metal surface. The asymmetric line shapes result from coupling of the photoelectrons to the conduction electrons.^[34] A multitude of factors have been identified as contributors to the line shape, particularly the full width at half maximum (FWHM), of the photoemission lines. These include phonon broadening, the lifetime of the photohole, the spectral resolution of the exciting x-ray beam and the energy resolution of the analyzer.^[30,34]

4 Experimental

4.1 Material

4.1.1 The Ag(100) crystal surface

A silver (Ag) single crystal that was provided by the team of the I09 beamline of the Diamond Light Source with a surface orientation of (100) was used as the substrate for the XPS experiments. It is important that a single crystal is used, as this has a uniform and precisely defined surface and has almost no structural defects.

Silver forms a face-centered cubic (fcc) crystal system, which is shown in Figure 4.1. This corresponds to the Pearson symbol cF. In the crystal system of silver, one silver atom is surrounded by twelve next neighbor atoms. Furthermore, the Ag(100) surface does not reconstruct. The lattice constant of silver in the volume crystal is $4.0853 \pm 0.0013 \text{ \AA}$ at a temperature of $23 \pm 3 \text{ }^\circ\text{C}$.^[35]

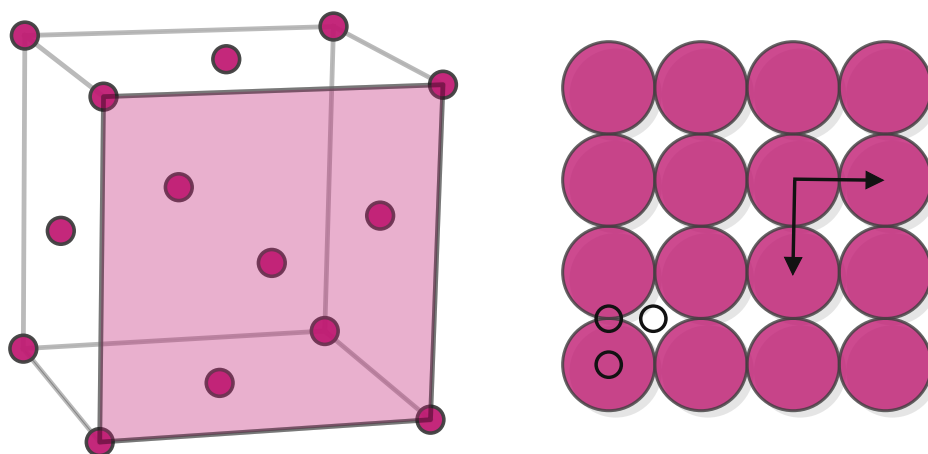


Figure 4.1: Unit cell of the silver crystal with marked (100) plane and the Ag(100) surface with the unit cell vectors (black arrows) and the distinct adsorption sites (black circles).

As seen in Figure 4.1, the unit cell of the Ag(100) surface is quadratic with a lattice constant $a_1 = a_2 = 4.0853 \text{ \AA} / \sqrt{2} = 2.8887 \text{ \AA}$. The Ag(100) surface has different adsorption sites called fcc, on-top and bridge. These sites are indicated by the black circles in Figure 4.1.

4.2 Experimental setup

All XPS measurements were performed in an ultra-high vacuum (UHV) chamber at the I09 endstation at the Diamond Light Source in Didcot, UK. A schematic overview of the I09 endstation is depicted in Figure 4.2.

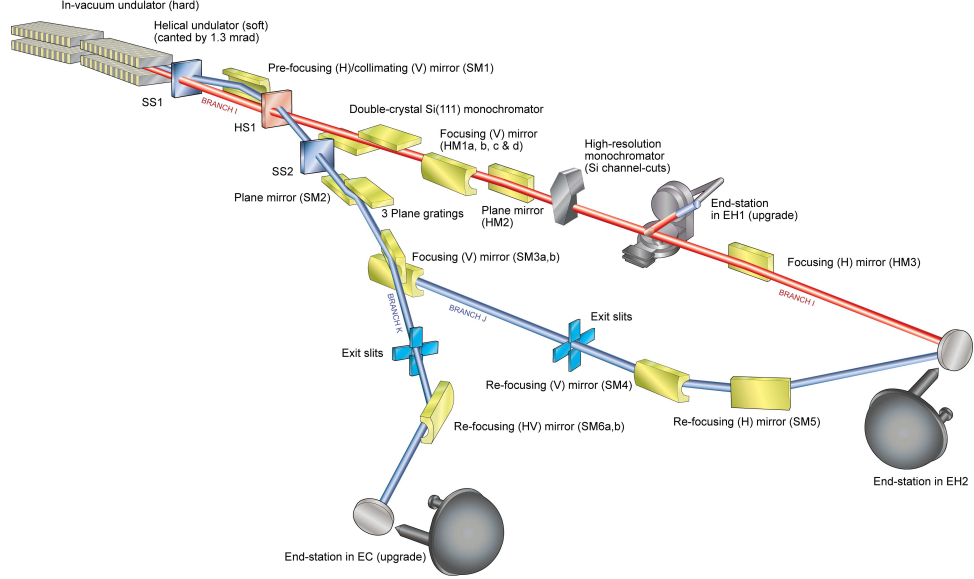


Figure 4.2: Schematic drawing of the I09 endstation of the Diamond Light Source in Didcot, UK. Picture taken from reference [36].

This particular endstation for the reported experiments (in EH2, Figure 4.2) utilizes radiation from an undulator located inside the electron storage ring of the synchrotron. The I09 endstation is equipped with a nitrogen-cooled Si(111) double crystal monochromator (hard x-rays) and a grating monochromator (soft x-rays). This setup enables high-intensity and precise x-ray measurements in the soft and hard x-ray ranges. The emitted photoelectrons were detected by a hemispherical EW4000 HAXPES analyzer purchased from Scienta Omicron. The analyzer has an acceptance cone of 56° and was mounted in the photon polarization plane.^[36] A schematic drawing of the sample position for the XPS measurements is illustrated in Figure 4.3.

The experimental sample preparation is described in the following. First, the Ag(100) surface was cleaned by sputtering with Ar^+ ions. Therefore, argon was dosed into the chamber up to a pressure of $5 \cdot 10^{-5}$ mbar and Ar^+ ions were accelerated with a voltage of 1 keV. Afterwards, the surface was heated up to 825 K for 45 min to increase the thermal diffusion and heal defects on the Ag(100) surface.

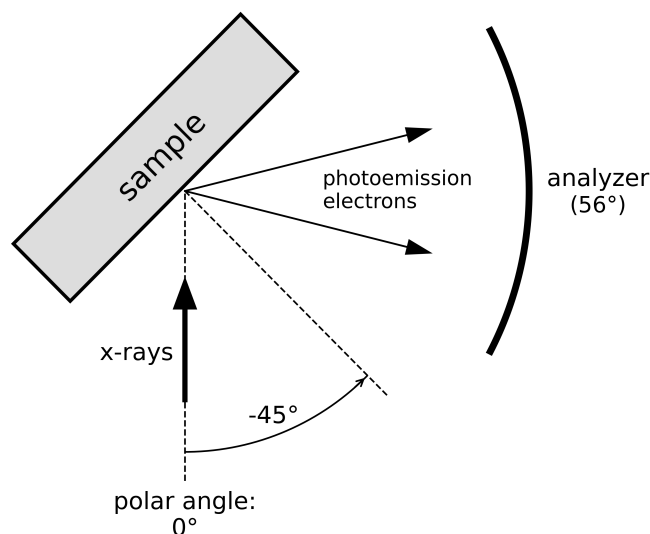


Figure 4.3: Top view of the sample geometries for the XPS measurements at the I09 endstation of the Diamond Light Source in Didcot, UK. Illustrated as in reference [37].

Subsequently, QA was evaporated on the Ag(100) surface with the sample being at room temperature. Therefore, the evaporator was heated to about 500 °C. The deposition process was performed according to procedures derived in the lab in Bonn, where the deposition could be controlled by a mass spectrometer and SPA-LEED measurements.

After evaporation, the phase of the QA molecules on the Ag(100) surface was characterized using LEED. Exemplary LEED pattern of the α - and the β -phase are shown in Figure 8.4. The α -phase shows a POL type superstructure with its characteristic streaky spots in the diffraction pattern, while the β -phase is commensurate with the Ag(100) surface and exhibits sharp spots in the diffraction pattern.

For the measurements of the XPS spectra, x-ray radiation with an energy of either 500 eV for C1s and N1s or 630 eV for O1s was used. For data collection, the sample normal was rotated by -45° out of the x-ray beam in the polar direction towards the analyzer.

The detailed parameters for the data acquisition are listed in Table 4.1. The pass energy E_{pass} of the analyzer determines the kinetic energy of the photoelectrons and therefore the energy resolution of the measurement. The step width is the energy difference between two consecutive data points in the XPS spectrum. The number of frames is the number of times the same spectrum was measured and averaged to reduce noise in the data.

Table 4.1: Experimental parameters for XPS measurements at the I09 endstation of the Diamond Light Source in Didcot, UK.

	photon energy E_γ	pass energy E_{pass}	step width	number of frames
C1s	500 eV	20 eV	40 meV	17
O1s	630 eV	20 eV	40 meV	17
N1s	500 eV	20 eV	40 meV	17

The parameters for the measurements were chosen to achieve a good signal-to-noise ratio and a high energy resolution. Furthermore, the acquisition time for each spectrum was as short as possible to prevent beam damage of the sample.

XPS spectra for the Fermi energy E_F were measured after each XPS measurement of C1s, O1s and N1s for calibrating the BE of the measured spectra against the Fermi level. E_F was measured with the same photon energy as the corresponding orbital and a pass energy of 20 eV. The step width was set to 40 meV and the number of frames was changed for each measurement.

4.3 Data processing

The measured data were then subjected to an initial processing stage, whereby the values from multiple datasets employing identical settings for the measurements were averaged to obtain an increased signal-to-noise ratio. The number of averaged spectra was different for the various preparations. Subsequently, the data underwent further processing with the software CasaXPS.^[38] Each XPS spectrum was processed in a consistent manner, as outlined below.

Initially, the BE of the XPS spectrum was calibrated against the Fermi energy E_F . For this purpose, an XPS spectrum of the Fermi edge with the same photon energy as the corresponding component was fitted with a Fermi function. Without inaccuracies of the devices, E_F should be zero. Accordingly, the binding energy was shifted by the value of the Fermi energy, which was determined from the fit of the Fermi function.

Afterwards, a linear background was fitted to the averaged and calibrated data. This background was then subtracted for further data processing. Then, the corresponding fitting models from chapter 5 were applied. The different peaks with line shape and constraints from Table 4.2 were used. These settings were the same for each processed dataset. Subsequently, the processed data was exported from CasaXPS and illustrated using a customized python script, enabling the generation of various plots as outlined in chapter 5.

Table 4.2: Parameters used in CasaXPS^[38] for the different peaks of the XPS spectra of QA on Ag(100).

peak	line shape	area ratio
C _{arom}	LA(1,8,800)	10
C _{NH}	LA(1,8,800)	4
C _{CO}	LA(1,8,800)	4
C _O	LA(1,8,200)	2
O1	LA(1,8,400)	1 (for β -phase)
O1 _{sat}	LA(1,1,400)	-
O2	LA(1,8,400)	1 (for β -phase)
O2 _{sat}	LA(1,1,400)	-
N1	LA(1,8,400)	1 (for β -phase)
N1 _{damage}	LA(1,8,400)	-
N2	LA(1,8,400)	1 (for β -phase)

For the line shapes, the $LA(\alpha, \beta, m)$ function was used, which is a numerical convolution of a Lorentzian and a Gaussian. The parameters α and β define the asymmetry of the line shape, where α describes the left part and β the right part. The parameter m defines the width of the Gaussian.

The line shapes for the different components were slightly changed in order to observe the best description of the XPS spectra. In general, the objective was to maintain consistency in the lines to the greatest extent possible.

Furthermore, the areas of the peaks were constrained to a certain value in order to ensure that the areas of the peaks are consistent with the stoichiometry of QA. The area constraints for the C1s XPS spectra were the same for both phases, whereas the area constraints of the O1s and N1s XPS spectra differed between the two phases. This was due to the changed number of hydrogen bonds per molecule in the two phases, which led to a changed number of distinct oxygen and nitrogen atoms in the XPS spectra.

5 Results

The ensuing results will be presented under three topics: the α -phase (section 5.1), the phase transition from the α - to the β -phase (section 5.2) and the β -phase (section 5.3). A compendium of all recorded XPS spectra, organized according to the different photoemission lines, is available for perusal in chapter 8.

5.1 The α -Phase

The investigation of the α -phase was conducted on all three available photoemission lines: C1s, O1s and N1s. The ensuing discourse will meticulously examine the α -phase.

Therefore, the present chapter has been subdivided into discrete sections, each dedicated to a specific atom type. Within each section, an overview of all recorded XPS spectra for that particular atom type is provided, along with a more detailed view of a representative XPS spectrum for a monolayer. A comparison will be made between the monolayer and the multilayer XPS spectrum.

5.1.1 C1s spectra

Figure 5.1 presents a series of C1s XPS spectra of QA adsorbed on the Ag(100) surface, recorded at varying coverages ranging from 0.08 ML to 1 ML. All spectra exhibit a dominant peak in the region around 285 eV, which is consistent across all coverages. Furthermore, a slight shoulder towards lower BEs is visible. Towards higher BEs, the XPS spectra show a more complex pathway with at least one peak.

Despite the overall similarity in the spectra, particularly in peak position and general shape, there are notable differences in intensity and variations in peak broadening as a function of coverage. The most apparent trend is the systematic increase in signal intensity with increasing coverage, indicating a proportional growth in the number of photoemitting C atoms due to the accumulation of molecules.

The spectrum with the highest integrated intensity that was not a multilayer spectrum is set to 1 ML. The difference between the monolayer and multilayer spectrum was the significant change in the peak shape, as seen in Figure 5.3. The adsorbed layers of the other spectra were assigned accordingly using their integrated intensities. This classification of the coverage was used for all XPS spectra for the α -phase. From the C1s XPS spectra, it can be concluded that there are different peaks corresponding to the C atoms in QA. This will be discussed in more detail below.

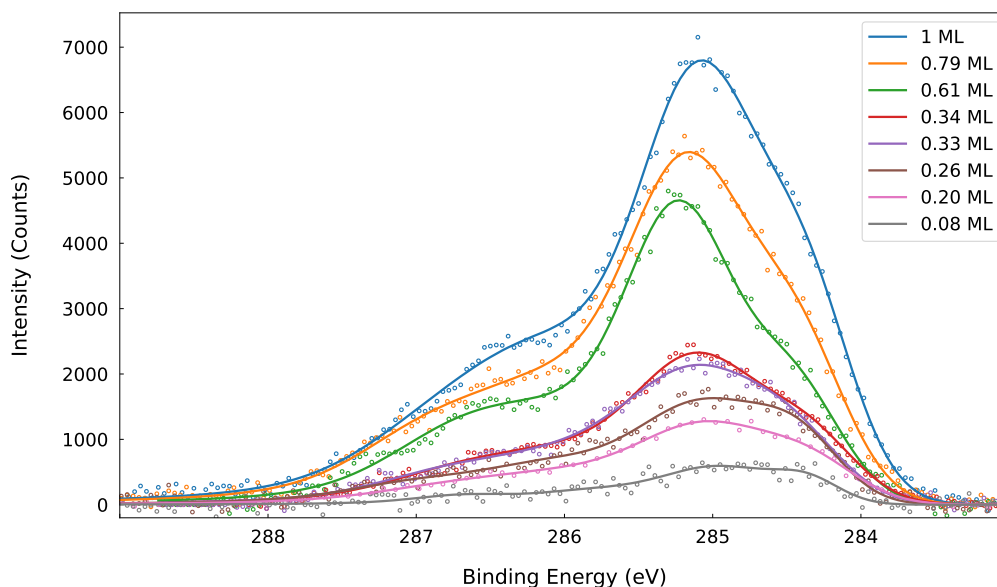


Figure 5.1: Overview of C1s XPS spectra for monolayer preparations of the α -phase of QA on Ag(100) with a coverage between 0.08 ML and 1.00 ML. The lines correspond to the envelope of the fitted peaks and the circles are the measured data points.

The XPS spectrum for a coverage of 1 ML in Figure 5.2 demonstrates different peaks that present C atoms with different chemical environments in QA.

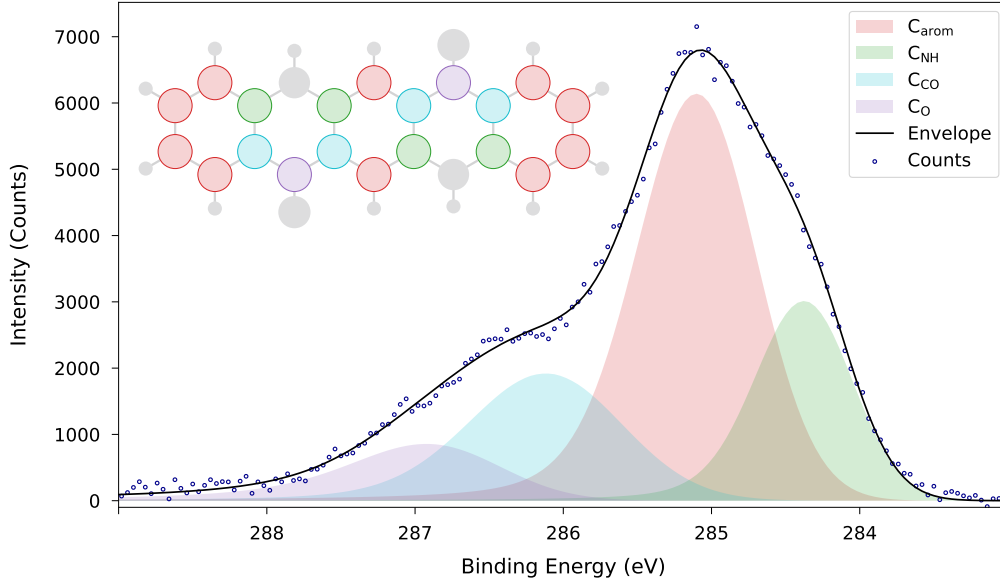


Figure 5.2: C1s XPS spectrum for the α -phase of QA on Ag(100) for a coverage of 1 ML with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line). Four C1s XPS spectra were averaged to obtain this spectrum.

The XPS spectra for the C1s photoemission lines of QA on the Ag(100) surface reveal consistent spectral features. The main peak (C_{arom}) contains contributions of the C atoms from the aromatic rings that are not part of the two rings containing the keto and the amine groups. Next, there is a peak for the neighbor C atoms of the amine (C_{NH}) and another peak for the neighbor C atoms of the ketone (C_{CO}). Finally, there is one peak for the carbonyl C atoms of the ketone (C_{O}). The values corresponding to each peak can be found in Table 5.1.

Table 5.1: Fit parameters used in CasaXPS^[38] for the monolayer α -phase of QA on Ag(100) for the C1s photoemission lines.

peak	BE / eV	area ratio	FWHM / eV
C_{arom}	285.02	10	0.94
C_{NH}	284.31	4	0.76
C_{CO}	286.01	4	1.20
C_{O}	286.72	2	1.15

As illustrated in Figure 5.2 and Table 5.1, the detailed fitting model can be described as follows: The BE of the C_{arom} peak is measured at 285.02 eV, with a FWHM of 0.94 eV. The C_{NH} peak is next to the aforementioned peak, with a BE of 284.31 eV and a FWHM of 0.76 eV. The analysis of the data reveals that the C_{CO} peak is located to the left of the C_{arom} peak. The BE of the C_{CO} peak is 286.01 eV and its FWHM is 1.20 eV. The C_{O} peak is located to the left of this peak at a BE of 286.72 eV and a FWHM of 1.15 eV.

In contrast to the α -phase C1s XPS spectra of the monolayer, the multilayer spectrum in Figure 5.3 exhibits a different shape.

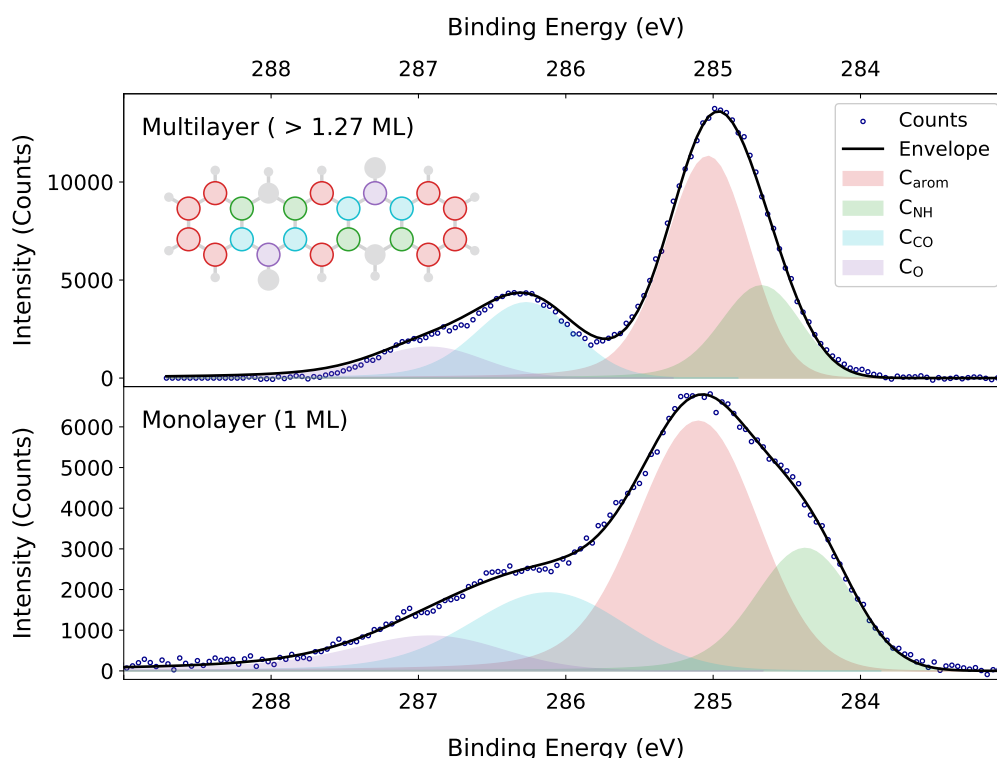


Figure 5.3: C1s XPS spectrum for the α -phase of QA on Ag(100) for a multilayer coverage compared to the XPS spectrum for a coverage of 1 ML with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line).

The coverage for the multilayer spectrum in Figure 5.3 was estimated to be above 1.27 ML. The coverage was determined by the integrated intensity of the C1s multilayer XPS spectrum, which was compared to the integrated intensity of the 1 ML XPS spectrum. Nevertheless, the exact coverage is not known, as the intensity of the photoemission lines of the multilayer is not directly proportional to the number

of adsorbed molecules, because the sensitivity of the photoemission lines changes for different layers. This leads to the assumption that the coverage was above 1.27 ML.

Initially, the multilayer spectrum is still adequately characterized by using the four distinct peaks, which possess equivalent area ratios to those of the monolayer spectrum. However, it is evident that the multilayer spectrum exhibits a significantly higher intensity relative to the monolayer spectrum. The primary distinctions are evident in the BE and the FWHM of the peaks. The spectrum displays a discernible minimum between the C_{arom} peak and the C_{O} peak. The peak associated with the amine group (C_{NH}) exhibits a shift towards the C_{arom} peak.

The parameters for the multilayer spectrum in Figure 5.3 are listed in Table 5.2.

Table 5.2: Fit parameters used in CasaXPS^[38] for the multilayer α -phase of QA on Ag(100) for the C1s photoemission lines.

peak	BE / eV	area ratio	FWHM / eV
C_{arom}	284.98	10	0.65
C_{NH}	284.62	4	0.62
C_{CO}	286.20	4	0.76
C_{O}	286.80	2	0.85

5.1.2 O1s spectra

Figure 5.4 presents a series of XPS spectra for the O1s photoemission lines of QA at three different coverages: 0.33 ML, 0.79 ML and 1 ML. A prominent peak centered near 531 eV is exhibited by all spectra, a characteristic of the O atoms in carbonyl functional groups present in the QA molecule. Additionally, the shape of the peak suggests the presence of a more intricate structure than a single peak. The spectra exhibit a comparable overall shape and peak position across the various surface coverages.

As anticipated, the spectral intensity increases with coverage, indicative of an increased number of QA molecules adsorbed on the surface. Beyond the intensity changes, differences in the peak width and asymmetry are also observed. The spectrum at the lowest coverage (0.33 ML) is narrower and more symmetric, whereas the spectra at higher coverages become progressively broader and more asymmetric.

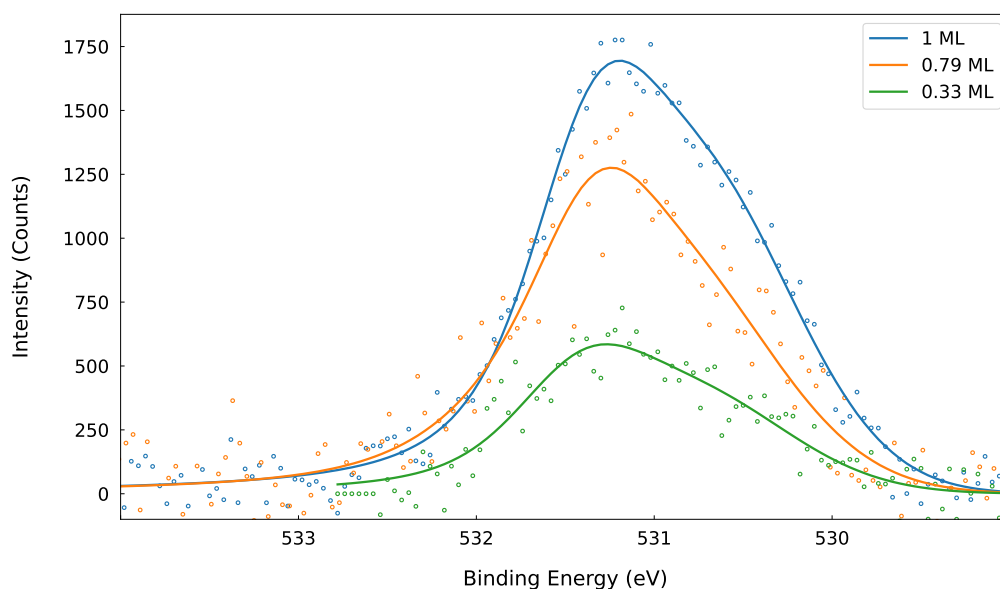


Figure 5.4: Overview of O1s XPS spectra for monolayer preparation of the α -phase of QA on Ag(100) with a coverage between 0.33 ML and 1.00 ML. The line corresponds to the envelope of the fitted peaks and the circles are the measured data points.

The O1s XPS spectrum of QA on Ag(100) at a 1 ML coverage reveals two distinguishable spectral components, which is shown in Figure 5.5.

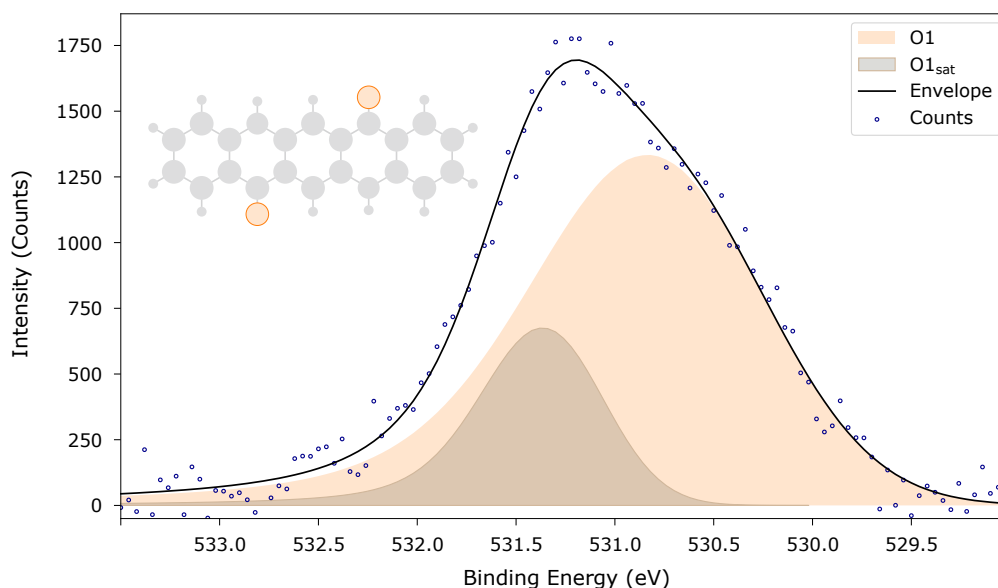


Figure 5.5: O1s XPS spectrum for the α -phase of QA on Ag(100) for a coverage of 1 ML with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line). Five O1s XPS spectra were averaged to obtain this spectrum.

As illustrated in Figure 5.5 and Table 5.3, the primary component (O1) is at a BE of 530.66 eV, exhibiting a FWHM of 1.36 eV. This peak is attributed to the core-level photoemission from O atoms situated within carbonyl functionalities. The relatively broad linewidth in comparison to the linewidths of the C1s XPS spectra is indicative of slight variations in the local chemical environment of these O atoms.

A secondary, less intense component (O1_{sat}) is observed at a higher BE of 531.27 eV with a narrower FWHM of 0.71 eV. This feature is interpreted as a shake-up satellite, arising from e.g. a $\pi - \pi^*$ transition within the conjugated π system of the QA molecule.

Quantitative analysis of the peak areas reveals that the satellite-to-main peak area ratio is approximately 1 : 0.27, which is equivalent to 21 % of the total O1s intensity. This ratio is consistent with the expected intensity of shake-up features in aromatic systems with significant conjugation.^[39]

The combination of these two components provides a comprehensive representation of the O1s photoemission lines of QA. This representation reflects both the intrinsic chemical identity of the O atoms and the dynamic electron processes that accompany core-level photoemission.

Table 5.3: Fit parameters used in CasaXPS^[38] for the monolayer α -phase of QA on Ag(100) for the O1s photoemission lines.

peak	BE / eV	area ratio	FWHM / eV
O1	530.66	2.00	1.36
O1 _{sat}	531.27	0.53	0.71

In Figure 5.6, a XPS spectrum with a multilayer coverage is compared to the previously discussed XPS spectrum.

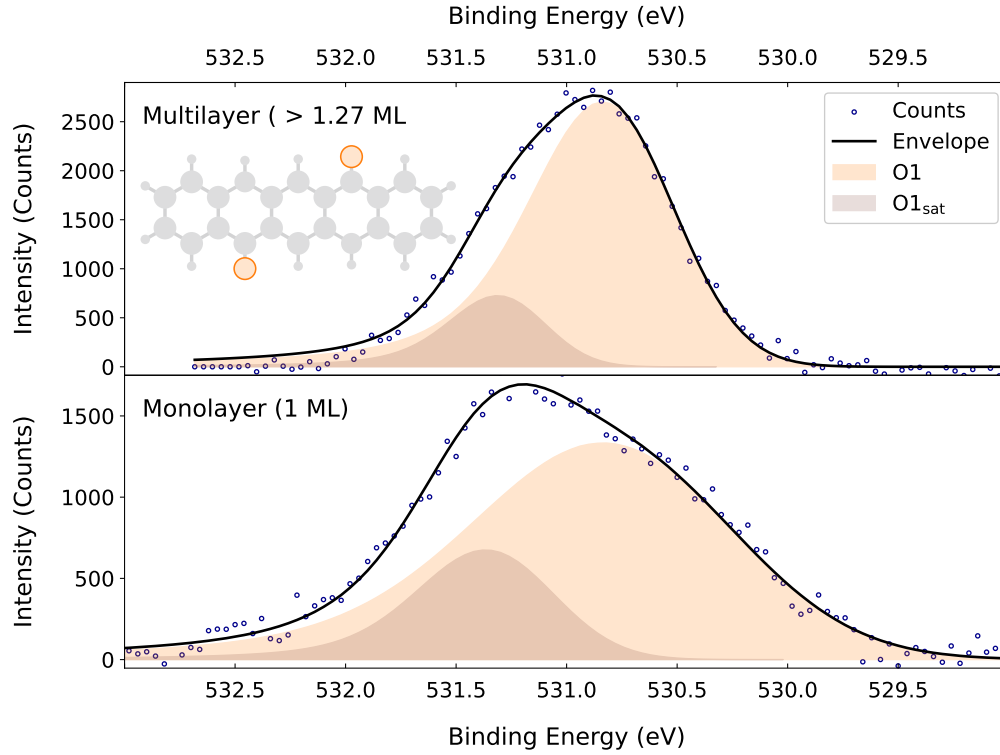


Figure 5.6: O1s XPS spectrum for the α -phase of QA on Ag(100) for a multilayer coverage compared to the XPS spectrum for a coverage of 1 ML with the experimental data points, the fitted components and the resulting envelope.

The O1s XPS spectrum recorded for QA on Ag(100) at multilayer coverage reveals a well-defined structure characterized by two distinct photoemission peaks. The primary component (O1) is at a BE of 530.733 eV and a FWHM of 0.75 eV. In comparison with the ML O1 peak, the peak position undergoes a change of approximately 0.076 eV.

A more salient finding is the increase in the FWHM of approximately 0.61 eV, which nearly corresponds to a 50 %.

The satellite peak ($O1_{\text{sat}}$) is observed at 531.24 eV with a FWHM of 0.5 eV. Once more, the position of the $O1_{\text{sat}}$ peak in relation to the ML remains constant. However, the FWHM of the peak decreases by 0.21 eV, which corresponds to approximately 30 %. This alteration may be attributable to the enhanced intensity exhibited by the multilayer XPS spectrum. The satellite-to-main ratio experiences a decrease from 1 : 0.27 in the ML to 1 : 0.18 in the multilayer, indicating a reduced shake-up process.

The parameters for the O1s XPS spectrum with a multilayer coverage in Figure 5.6 are listed in Table 5.4.

Table 5.4: Fit parameters used in CasaXPS^[38] for the multilayer α -phase of QA on Ag(100) for the O1s photoemission lines..

peak	BE / eV	area ratio	FWHM / eV
O1	530.73	2.00	0.75
$O1_{\text{sat}}$	531.24	0.36	0.50

5.1.3 N1s spectra

The XPS spectra displayed in Figure 5.7 show the N1s photoemission lines of QA on Ag(100) with varying coverages between 0.26 ML and 1 ML. A prominent main peak centered around 400.1 eV is exhibited by all spectra. Despite the general spectral similarities across different coverages, clear distinctions emerge in the shape and intensity of the main peak. As the coverage increases from submonolayer to monolayer, the main peak systematically increases in intensity, reflecting the proportional increase in nitrogen-containing molecules. In scenarios where coverages are less substantial, the spectra tend to manifest as marginally broader. Conversely, as the coverage increases, the spectra become more narrow.

Additionally, a secondary peak is evident on the low-BE side of the main peak in some spectra at approximately 397.7 eV. In summary, while the core features of the XPS of QA remain largely preserved across different coverages, changes are observed in the peak intensity and the width, in addition to the second peak at a lower BE.

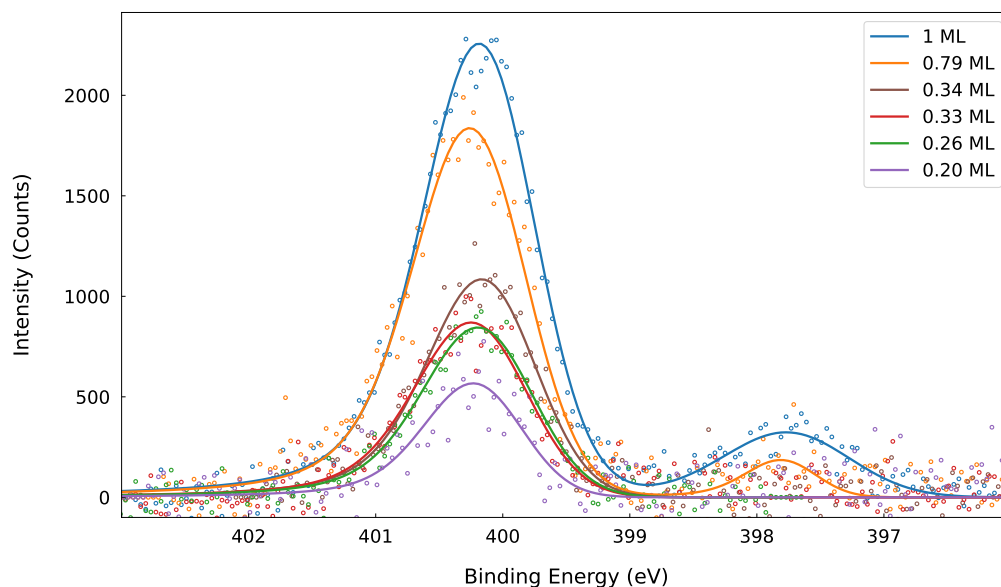


Figure 5.7: Overview of N1s XPS spectra for monolayer preparation of the α -phase of QA on Ag(100) with a coverage between 0.26 ML and 1.00 ML. The line corresponds to the envelope of the fitted peaks and the circles are the measured data points.

The N1s XPS spectrum of QA on Ag(100) with a coverage of 1 ML, as depicted in Figure 5.7, is discussed in more detail in Figure 5.8, which presents the different peaks.

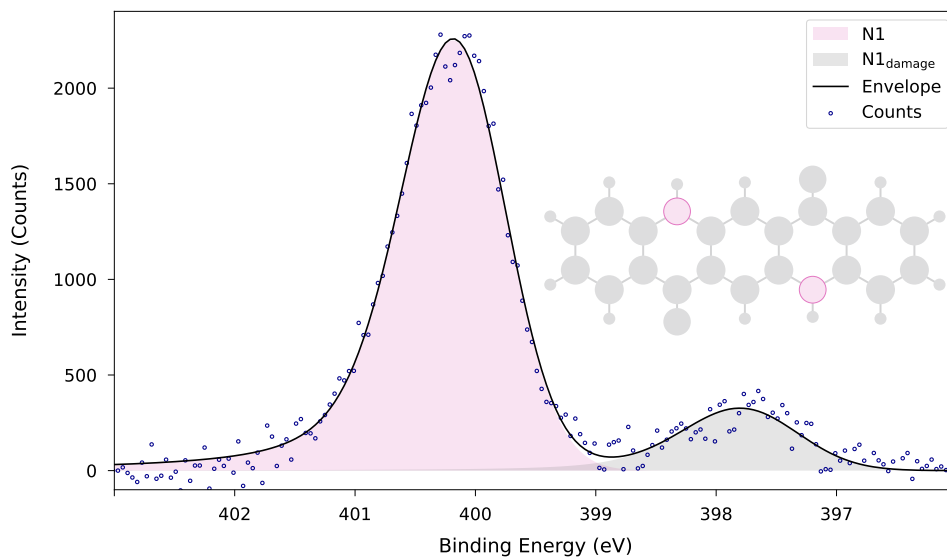


Figure 5.8: N1s XPS spectrum for the α -phase of QA on Ag(100) for a coverage of 1 ML with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line). Four N1s XPS spectra were averaged to obtain this spectrum.

The XPS spectrum in Figure 5.8 can be divided into two discrete components: a primary peak (N1) and a secondary feature at lower BE (N1_{damage}). The precise values for the two peaks are enumerated in Table 5.5. The N1 peak, with a center at 400.05 eV and a FWHM of 1.02 eV, is attributed to N atoms embedded within the chemically intact QA molecule. The narrow FWHM and symmetrical peak profile are indicative of a well-defined and electronically homogenous nitrogen environment.

The secondary component, N1_{damage}, emerges at a BE of 397.66 eV and exhibits a FWHM of 1.16 eV. This phenomenon is attributed to the presence of chemically modified nitrogen species, which are most likely the result of radiation-induced damage due to prolonged x-ray exposure. The assumption that the N atoms in QA are susceptible to radiation damage is supported by the XPS spectra for 1 ML and 0.79 ML coverages, which exhibit a different area ratio of the main component to the damage component. These two XPS spectra were longer exposed to the x-ray radiation than the other XPS spectra without a second peak, explaining the observed peak due to beam damage.

A thorough qualitative analysis was conducted, which yielded a damage-intensity-ratio of 0.16 for the 1 ML coverage. This ratio indicated that the majority of N atoms maintain their molecular integrity under the measurement conditions, while a minor but significant portion experiences beam damage.

Table 5.5: Fit parameters used in CasaXPS^[38] for the monolayer α -phase of QA on Ag(100) for the N1s photoemission lines.

peak	BE / eV	area ratio	FWHM / eV
N1	400.05	2.00	1.02
N1 _{damage}	397.66	0.31	1.16

A comparison of the N1s XPS spectrum of the monolayer and the spectrum corresponding to the multilayer is illustrated in Figure 5.9.

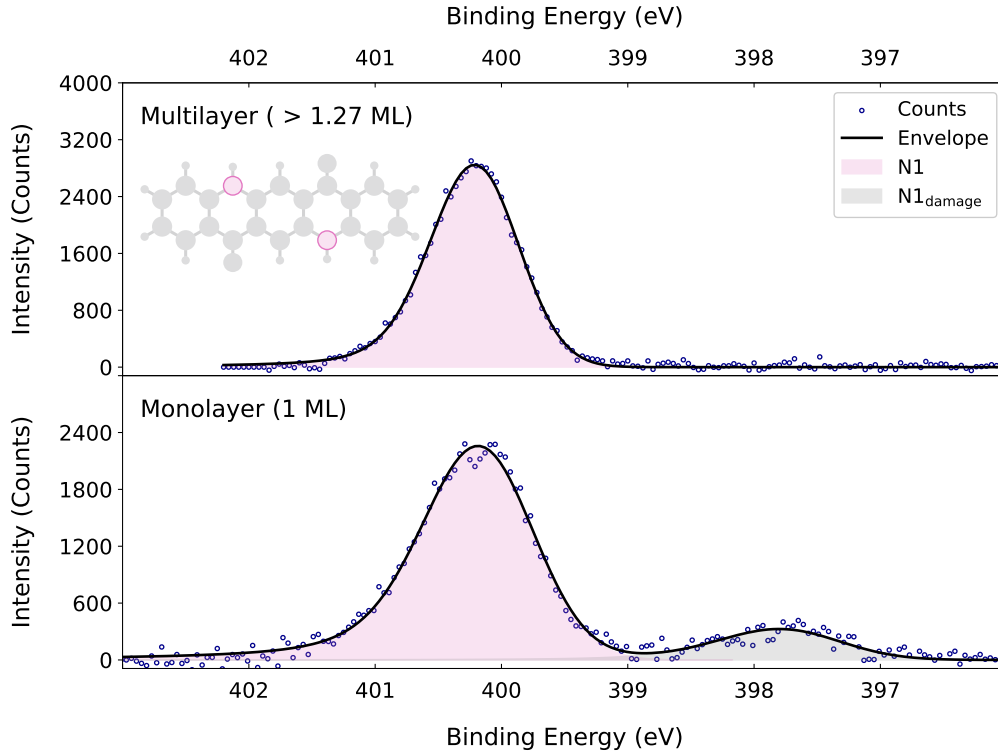


Figure 5.9: N1s XPS spectrum for the α -phase of QA on Ag(100) for a multilayer coverage compared to the XPS spectrum for a coverage of 1 ML with the experimental data points, the fitted components and the resulting envelope.

The XPS spectrum with a multilayer coverage reveals a single component (N1 centered at a BE of 400.15 eV, accompanied by a FWHM of 0.82 eV. This observation signifies a subtle modification in the chemical environment surrounding the N atom. The peak associated with beam damage (N1_{damage}) remains undetected. This absence indicated that the spectrum of the multilayer originated predominantly from chemically intact

nitrogen species. The BE exhibits a slight increase (0.09 eV) in the multilayer, accompanied by a narrower FWHM.

The parameters for the N1s XPS spectrum with a multilayer coverage in Figure 5.9 are listed in Table 5.6.

Table 5.6: Fit parameters used in CasaXPS^[38] for the multilayer α -phase of QA on Ag(100) for the N1s photoemission lines.

peak	BE / eV	area ratio	FWHM / eV
N1	400.15	2.00	0.82

5.2 C1s XPS spectra for the phase transition

The Figure 5.10 illustrates C1s photoemission lines for QA molecules absorbed on the Ag(100) surface, showing the transition from the α - to the β -phase with a coverage of 0.58 ML. The XPS spectra are arranged vertically, with the α -phase shown at the top, the β -phase at the bottom and a series of intermediate XPS spectra representing the thermally induced phase transition. Maximum values for each individual peak are denoted by a colored vertical line. The phase transition occurs at a temperature of 450 K, which was the target temperature during the recording of the XPS spectra. This temperature was chosen based on the previous results of the LEED measurements, which indicated that the phase transition from the α - to the β -phase occurs at approximately 450 K.

In the α -phase, the C1s XPS spectrum is characterized by distinct peaks corresponding to various chemically different C atoms, including the C_{arom} , C_{NH} , C_{CO} and C_{O} peaks, as discussed in subsection 5.1.1. In the final β -phase, the C1s XPS spectrum demonstrates altered peak intensities and BE in comparison to the initial α -phase. The same fit model was used for both phases, but the BE and FWHM of the peaks were allowed to change during the fitting procedure.

It is noteworthy that all peaks undergo modification, which is indicative of a thermally induced structural rearrangement or reorientation of the QA molecule. The sequence demonstrates a substantial trend for all peaks. A systematic, stepwise decrease in the BE is exhibited by all four peaks. A comparative analysis of the C_{arom} , C_{CO} and C_{O} peaks reveals a consistent pattern of change.

A further difference in the change is observed in the C_{NH} peak. The shifts towards lower BEs is less pronounced in comparison to the other peaks. This indicates that the environments of the other C1s photoemission lines, namely C_{arom} , C_{CO} and C_{O} , are changed in a similar way, while the C_{NH} C atoms experiences a different change in its chemical surrounding.

The analysis of the peak areas revealed that the integrated area of the peaks decreases by 13 % from the α - to the β -phase. It could be possible that this change is due to the different orientation of the QA molecules in the β -phase, but this is not certain. Additionally, the number of QA molecules in the β -phase may be different from the number of QA molecules in the α -phase due to desorption of some QA molecules during the thermal induced phase transition.

The time between the α -phase XPS spectrum and the β -phase XPS spectrum was 49 minutes. As the first XPS spectrum of the α -phase was recorded, the temperature was set to 450 K, initiating the heating of the sample. The sample temperature was 450 K for the β -phase XPS spectrum. The exact temperature for the intermediate XPS spectra are not known.

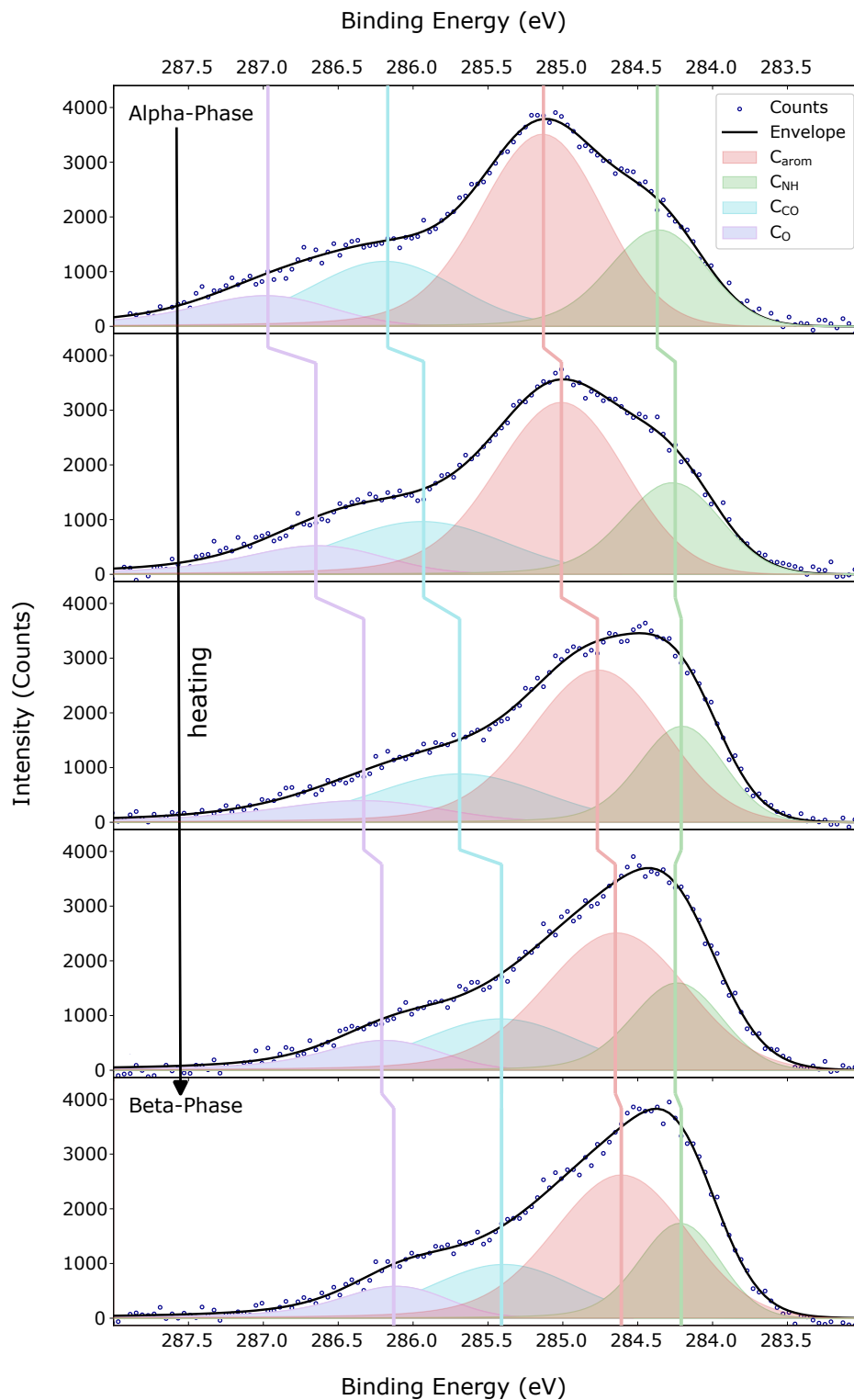


Figure 5.10: C1s XPS spectrum for the phase transition from the α - to the β -phase of QA on Ag(100) with a coverage of 0.58 ML. Integrated area of the peaks decreases by 13 % from the α - to the β -phase. The vertical lines indicate the maximum of each peak.

5.3 The commensurate β -Phase

The ensuing discourse will examine the phase designated as the β -phase. The investigation of the β -phase was conducted once more on all three available photoemission lines: C1s, O1s, and N1s. To address this need, the present chapter has been meticulously subdivided into discrete sections, each dedicated to specific photoemission lines. A comprehensive analysis of a particular XPS spectrum of the β -phase is presented in each section, along with a comparative analysis of the α - and β -phases.

5.3.1 C1s spectra

The C1s XPS spectrum for the β -phase of QA on Ag(100) is presented in Figure 5.11.

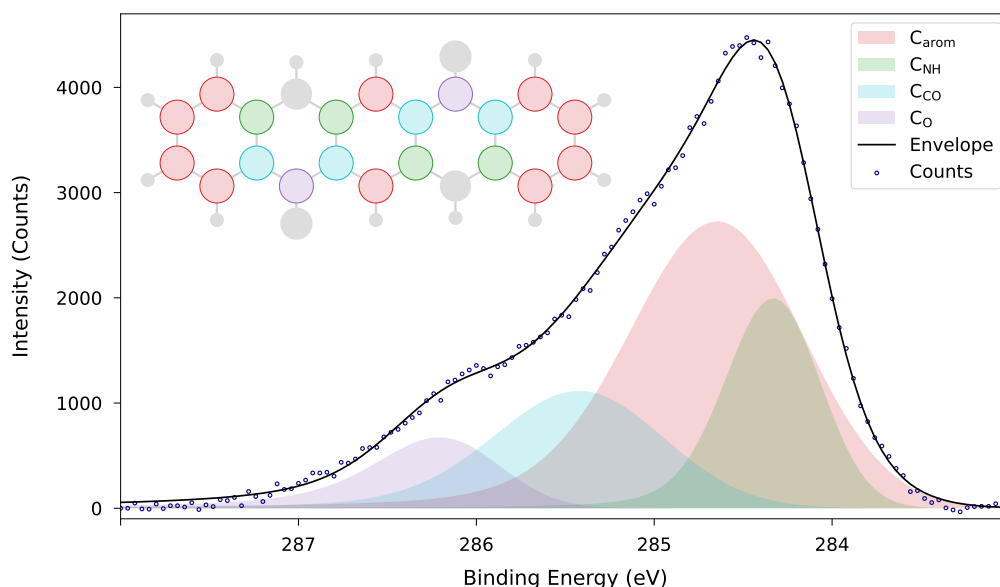


Figure 5.11: C1s XPS spectrum for the β -phase of QA on Ag(100) for a coverage of 0.54 ML with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line).

The XPS spectrum of QA in the β -phase displays four distinct components, designated as C_{arom} , C_{NH} , C_{CO} and C_{O} , which bear the same nomenclature as in the α -phase. This finding suggests that the various C atoms present within the QA molecule exhibit a persistent correlation with distinct peaks in the XPS spectrum.

The primary component, C_{arom} , is located at a BE of 284.55 eV with a FWHM of 1.13 eV. The second peak, characterized by a BE of 284.28 eV and a FWHM of

0.62 eV, is identified as the C_{NH} peak. The C_{CO} peak is located at a BE of 285.32 eV, experiencing a FWHM of 1.11 eV. In the vicinity of this peak, at a higher BE, is the C_O peak at 286.07 eV with a FWHM of 0.79 eV.

A qualitative investigation of the peak areas yielded analogous results to those of the α -phase, wherein the area ratios were set to the number of C atoms in the QA molecule.

In summary, the XPS spectrum for the C1s components of the β -phase offers significant insights into the chemical changes that occur between the α - and β -phase. The single peaks are compared to each other, thereby providing information about the local environment around the C atoms.

Table 5.7: Fit model of the β -phase for C1s.

peak	BE / eV	area ratio	FWHM / eV
C_{arom}	284.55	10	1.13
C_{NH}	284.28	4	0.62
C_{CO}	285.32	4	1.11
C_O	286.07	2	0.79

In order to facilitate a more direct comparison of the β -phase to the α -phase, the C1s XPS spectra of these two phases are presented in Figure 5.12.

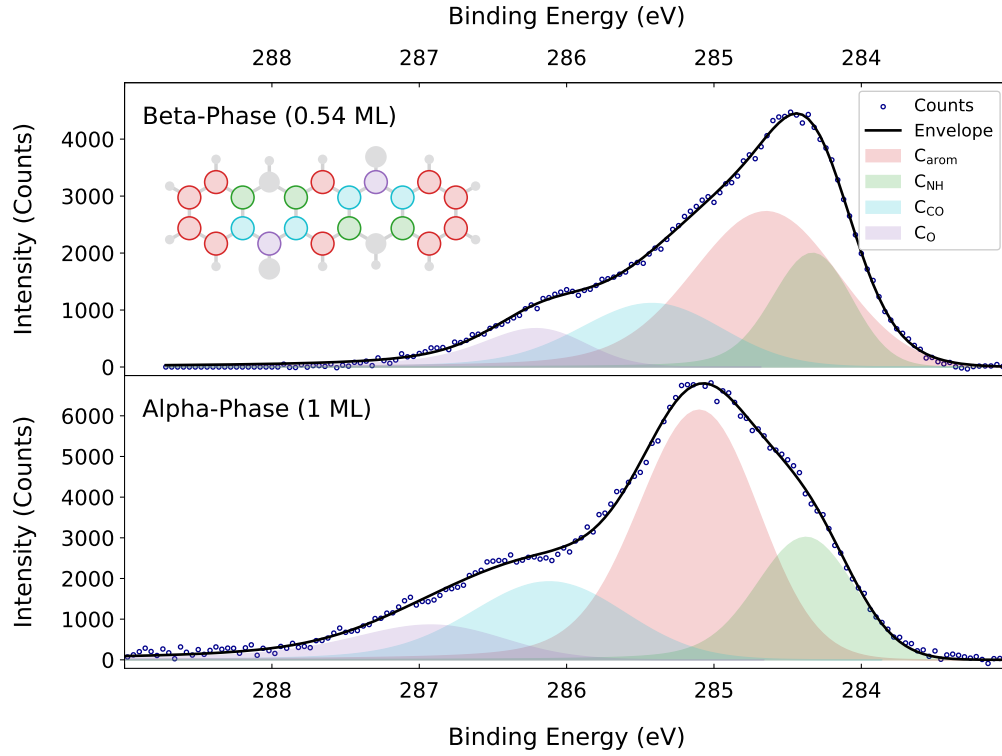


Figure 5.12: C1s XPS spectrum for the α - and β -phase of QA on Ag(100).

A close examination reveals that all four peaks shift to a lower BE. The shift in the BE of the C_{NH} peak is, by far, the least significant. The three other peaks, C_{arom} , C_{CO} and C_O , demonstrate a comparable shift in the BE.

A further noteworthy discrepancy is the increase of the FWHM for the C_{arom} peak, which is approximately 0.19 eV. The C_{NH} and C_{CO} peaks demonstrate a decrease in their FWHM by approximately 0.1 eV. It is remarkable that the FWHM of the C_O peak decreases by 0.37 eV. The observed variations of the FWHM could be attributed to the interdependence of all four peaks.

In summary, the alterations in the BE and the FWHM values of the four peaks result in a significantly different XPS spectrum. Consequently, QA on Ag(100) in the β -phase is expected to demonstrate a significant change in its structure compared to the α -phase structure.

5.3.2 O1s spectra

In Figure 5.13, the XPS spectrum for the O1s components for the β -phase of QA on Ag(100) is presented.

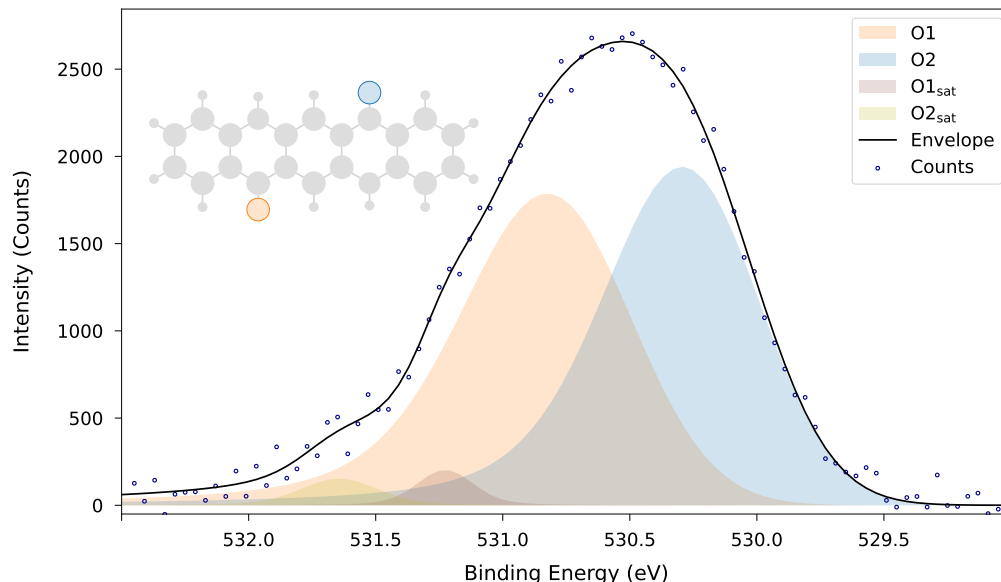


Figure 5.13: O1s XPS spectrum for the β -phase of QA on Ag(100) for a coverage of 0.54 ML with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line).

The XPS spectrum of the β -phase in Figure 5.13 consists of two peaks, each with its own satellite. The detailed data of the overall four peaks are shown in Table 5.8. The O1 peak is located at a BE of 530.68 eV with a FWHM of 0.71 eV. The respective satellite is at 531.66 eV with a FWHM of 0.36 eV. This corresponds to a shift in BE of 0.98 eV.

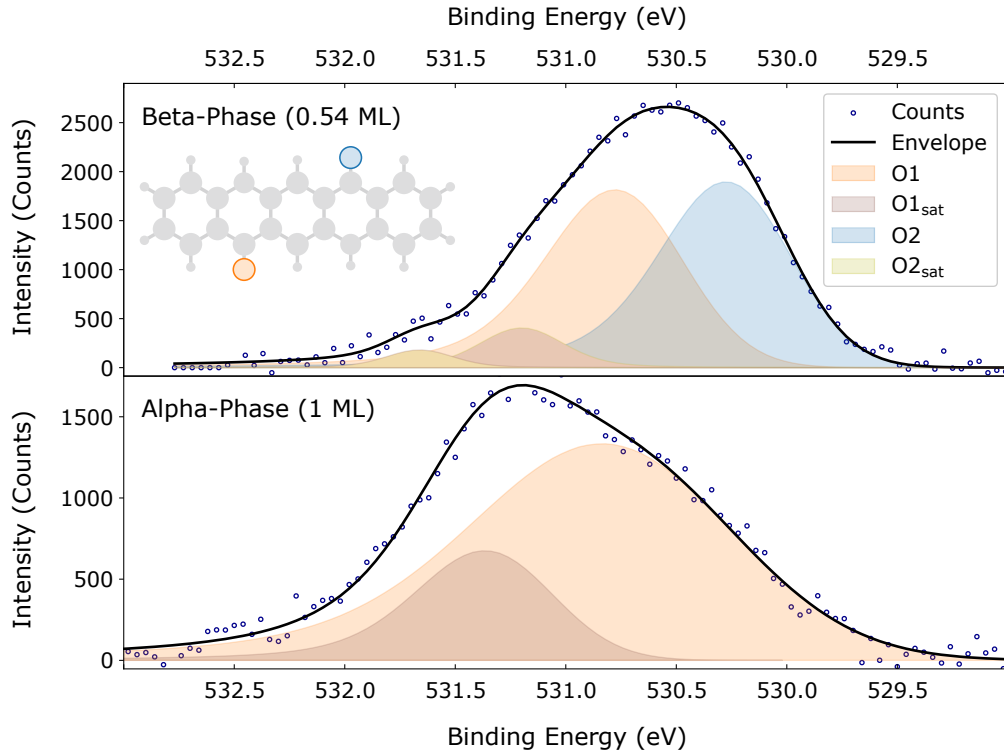
The other peak, O2, has a slightly lower BE of 530.19 eV but a similar FWHM of 0.68 eV. The satellite O2_{sat} also has a similar shift of 1.01 eV and is located at a BE of 531.20 eV with a FWHM of 0.45 eV. The O1 and O2 have an area ratio of 1:1, which corresponds to one O atom for each peak.

In summary, the FWHM of two peaks are around 0.7 eV and both have a shake-up satellite with a higher BE by approximately 1.0 eV. The two peaks are 0.5 eV apart from each other.

Table 5.8: Fit model of the β -phase for O1s.

peak	BE / eV	area ratio	FWHM / eV
O1	530.68	1	0.71
O1 _{sat}	531.20	0.14	0.45
O2	530.19	1	0.68
O2 _{sat}	531.66	0.05	0.36

To enable a better comparison between α - and β -phase, both spectra are depicted in Figure 5.14.

Figure 5.14: O1s XPS spectrum for the α - and β -phase of QA on Ag(100).

The investigation of the differences between the two phases yields that the O1 peak does not change its position but exhibits a change in its FWHM towards a smaller value. Same holds true for the O1_{sat} peak, which also changes its BE by approximately 0.4 eV.

The most significant change is the appearance of a second peak for the β -phase. This means that the two chemically equivalent peaks from the α -phase have changed their environment, leading to two chemically different O atoms in the β -phase.

5.3.3 N1s spectra

In Figure 5.15, the N1s XPS spectrum for the β -phase of QA on Ag(100) is presented.

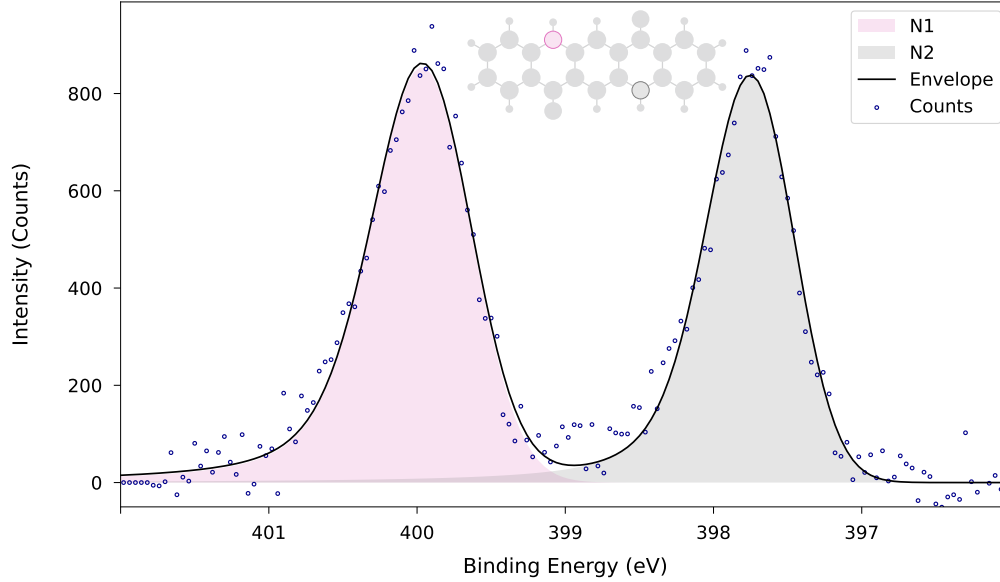


Figure 5.15: N1s XPS spectrum for the β -phase of QA on Ag(100) for a coverage of 0.54 ML with the experimental data points (circles), the fitted components (areas) and the resulting envelope (line).

Figure 5.15 and Table 5.9 enable a quantitative description of the N1s XPS spectrum for the β -phase. As seen, the β -phase has two different peaks for N1s, namely N1 and N2. Both peaks have a similar FWHM of 0.77 eV and 0.69 eV. Furthermore, they have the same intensities and an area-ratio of 1:1. This indicates that each peak corresponds to one of the N atoms of QA.

The main difference of the two peaks is their BE. The N1 peak is located at a BE of 399.87 eV while the N2 peak has a BE of 397.66 eV. This means that the peaks are separated by 2.20 eV. In comparison to the shifts in BEs for the other photoemission lines, this value seems to indicate a quite drastic change in the chemical environment of one of the N atoms.

Table 5.9: Fit model of the β -phase for N1s.

peak	BE / eV	area ratio	FWHM / eV
N1	399.87	1	0.77
N2	397.66	1	0.69

To enable a comparative analysis between the α - and the β -phase, the two phases are depicted in a juxtaposed arrangement in Figure 5.16.

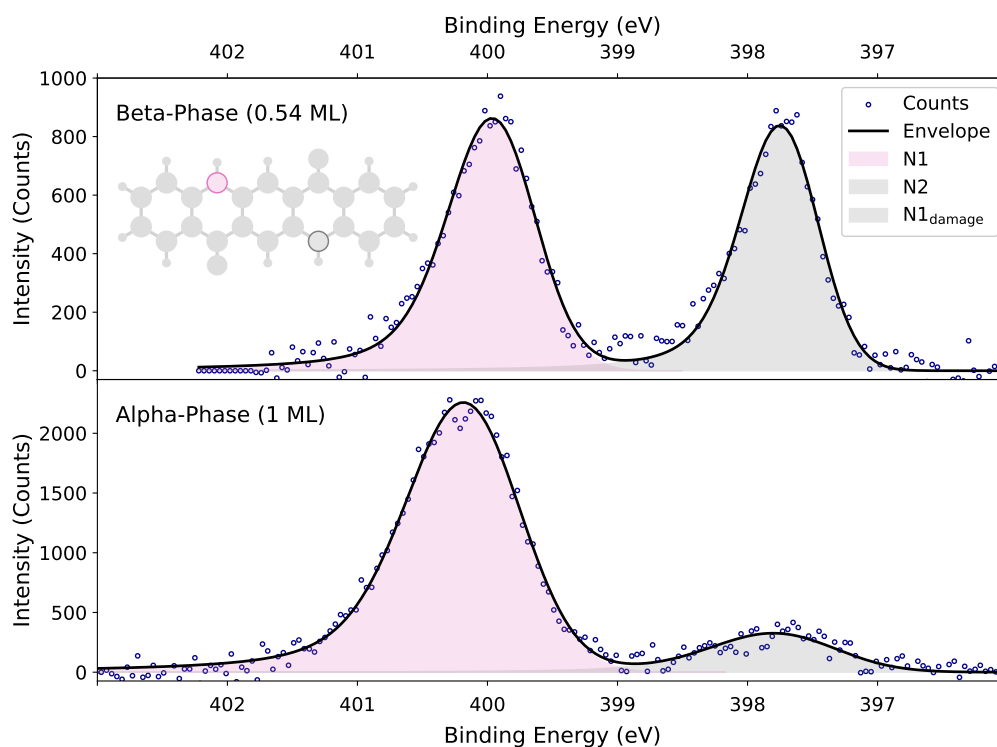


Figure 5.16: N1s XPS spectrum for the α - and β -phase of QA on Ag(100).

The comparison of the α - and the β -phase yields interesting results about the N atoms. First, the N1 peak keeps almost the same BE in both phases. There is only a change in BE of about 0.188 eV towards a lower value. The more significant change can be observed in the change of the FWHM, decreasing from 1.017 eV to 0.774 eV. This corresponds to a sharper peak in the XPS spectrum, indicating a more homogenous environment for the N1 peak.

Nevertheless, the other peak, namely N2, reveals a more intriguing finding, prompting further examination. The N2 peak has exactly the same BE as the N1_{damage} peak of the α -phase. As seen in Figure 5.16, the areas of the second peak in the different phases can not be compared anyhow, which makes an comparison of the FWHM unsuitable, because the N1_{damage} peak is due to beam damage and the N2 peak are a chemically different N atoms in the β -phase.

5.4 Binding energy changes upon the phase transition

For a further comparison of the differences in the BEs between the α - and the β -phase, an overview of these are illustrated in Figure 5.17.

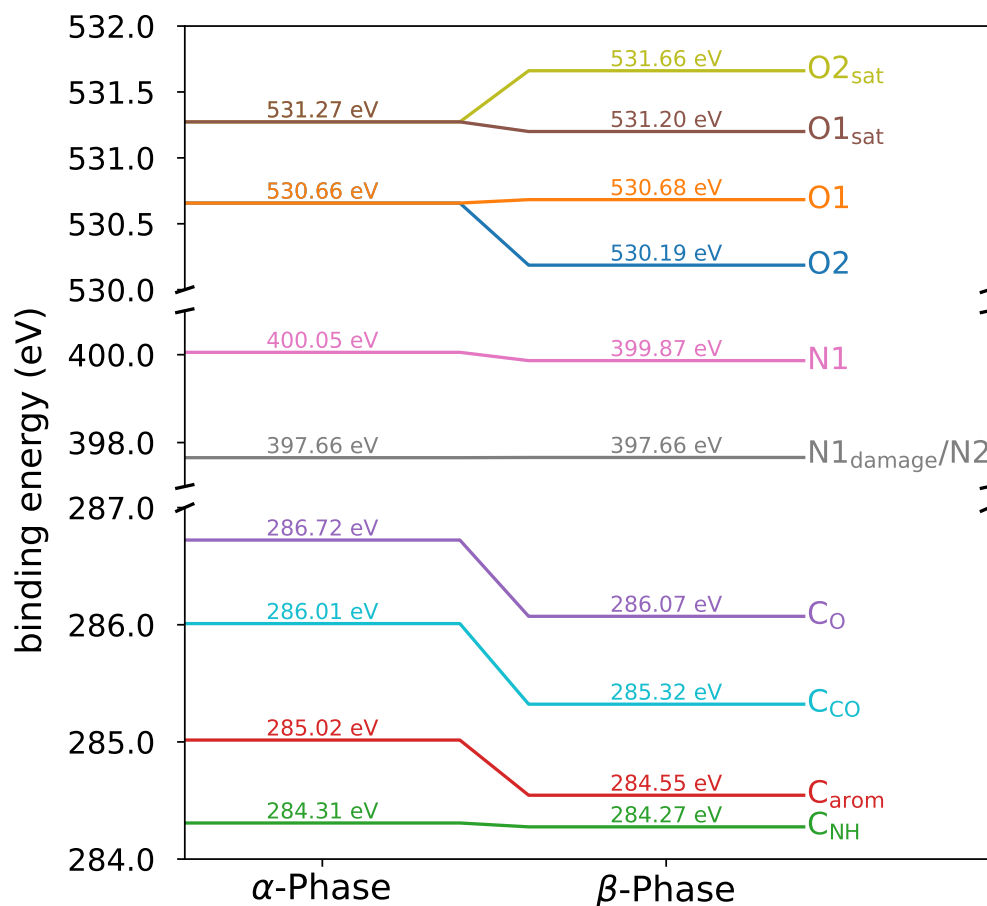


Figure 5.17: Illustration of the shift in the BE from the α - to the β -phase of QA on Ag(100) for the different peaks in the XPS spectra for all core-levels.

Initially, a significant trend is observed in all peaks across all XPS spectra. The BEs of all atoms in the β -phase are lower than or similar to those of the atoms in the α -phase.

Starting with the XPS spectra for C1s, the differences in the BEs are subsequently examined in detail. Shifts in the BE for three of the four peaks, namely C_{arom}, C_{CO} and C_O are similar. They all decrease by 0.5-0.7 eV. Conversely, the BE of the C_{NH}

peak does not significantly decrease. This indicates that the C atoms next to the amine undergo a different change during phase transition than the other C atoms.

Consequently, the XPS spectra for N1s are of great interest. As previously mentioned, the β -phase exhibits two peaks instead of one for N1s. The second peak in the β -phase is more than 2 eV lower in BE than the first peak. This represents a significant difference in the BE that is not observed for any other peak. To provide an example, the BE difference between the C_{NH} and the C_O peak, despite their very different chemical environments, is only 1.8 eV.

Lastly, the BEs of the two phases for the O1s XPS spectra are compared. Again, there are two distinct peaks in the β -phase instead of one in the α -phase. The BE of one peak remains at roughly the same value, while the other decreases by about 0.4 eV. This result is similar to the change in the BE of individual components observed in the C1s XPS spectra.

6 Discussion

The α - and β -phase of QA on the Ag(100) surface were investigated using XPS for three different components: C1s, O1s and N1s. The following discourse will examine the results obtained from chapter 5, which will be divided in the same manner as the results section.

6.1 The α -Phase

The C1s XPS spectra consist of four distinct components, corresponding to the distinct types of C atoms in the QA molecule, namely C_{arom} , C_{NH} , C_{CO} and C_{O} .

The C atoms C_{arom} have the least influence from the functional groups. This leads to a BE in the center of the spectrum. Due to the number of C atoms associated with this peak, the peak area is the most substantial, and the peak provides the primary contribution to the overall shape of the spectrum. The BE of the peak appears to be realistic, as evidenced by the agreement of the BE with that of other molecular adsorbates on the Ag(100) surface, such as perylenetetracarboxylic dianhydride (PTCDA) or indigo. For PTCDA, the BE of the C atoms inside the ring is 284.91 eV,^[39] which is very similar to the BE of the C_{arom} peak (285.02 eV) for QA. In addition, the BE of the corresponding peak in indigo (285.5 eV)^[40,41] is also very similar to the BE of the C_{arom} peak in QA.

The C_{CO} peak is located at a higher BE than the C_{arom} peak, which is expected due to the electronegative O atom. The BE of the C_{CO} peak (286.01 eV) is also in line with the literature, as it is similar to the BE of C atoms next to a carbonyl groups in other molecular adsorbates, such as PTCDA (285.47 eV).^[39]

The C_{O} peak is located at an even higher BE than the C_{CO} peak, which is also expected due to the electronegative O atom. The BE of the C_{O} peak (286.72 eV) also matches to the literature, as it is similar but not as high as the BE of carbonyl C atoms in PTCDA (288.92 eV)^[39] or in indigo (287.5 eV).^[40,41]

The peak corresponding to the C atoms adjacent to the amine, designated as C_{NH} , is located at a slightly decreased BE in comparison to the C_{arom} peak. This behavior of aromatic systems with amine functional groups inside, such as those found in pyridine, is typically observed.^[42,43] In contrast to this, the BE of the C_{NH} peak in indigo (285.9 eV)^[40] is higher than the BE of the C_{arom} peak, which is not the case

for QA. Nevertheless, the structures and the bonding situation of the two molecules are different, which could explain the different BEs.

The XPS spectra for the N1s component exhibit a single peak in the α -phase. The BE of the N1s peak (400.0 eV) is in line with the literature, as it is similar to the BE of N atoms in other molecular adsorbates, such as indigo (399.9 eV).^[44]

The peak that arises due to beam damage, N1_{damage}, has a shift towards smaller BEs of around 2.20 eV. This large difference indicates a drastic change in the chemical environment of the N atoms. Compared to other shifts in the BE, e.g. for the C atoms, the influence of other neighboring atoms are not enough to explain this change. For this reason, something different has to occur at the N atom due to beam damage.

A typical observed phenomena for the amine group is the deprotonation of the N atom. The deprotonation of amine groups in XPS spectra was examined with a shift in the BE of about 2-3 eV for glyphosate, molecules similar to pyridine and polyaniline films.^[45–47] This matches to the results found for the N1_{damage} peak, leading to the assumption that the x-ray beam deprotonates some of the N atoms.

Last, the XPS spectra for the O1s components are discussed. The BE of the O1 peak (530.66 eV) matches similar systems. e.g. PTCDA on the Ag(100) surface (531.93 eV)^[39] or indigo (530.2 eV).^[44]

A comparable satellite peak was also measured for PTCDA with a difference to the main component of 2.28 eV.^[39] The satellite peaks for the two adsorbates have similar shifts in their BEs and similar area ratios with respect to the main peak. This indicates that the results of the O1s XPS spectra and the fit model is realistic. Indigo exhibits a similar satellite peak, but with a shift of 1.0 eV.^[44]

Nevertheless, the photoemission lines for the O1s components should also exhibit a second peak, which corresponds to the O atom that cannot form a hydrogen bond with the amine group of another molecule because some of the N atoms are deprotonated due to beam damage. This peak is not observed in the XPS spectra, which confirms that the number of deprotonated N atoms is low and does not have a significant impact on the overall spectra for the α -phase. Nevertheless, a second peak for the O1s component could be hidden in the XPS spectra, increasing the FWHM of the O1 peak.

In summary, the entirety of the peaks in the XPS spectra adequately describe the molecular structure of QA. The BE of the different photoemission lines is consistent with the given values from the literature. In addition, the observed peaks in the XPS spectra corroborate the existing structure model proposed by N. Humberg.^[16] The chemical environments of all N and O atoms are similar within the structure model, because all O atoms of one molecule form a hydrogen bond with the amine group of another molecule. This is evident from the XPS spectra, which exhibit only a

single peak for the corresponding atoms. Furthermore, the results indicate that all molecules in the structure have a similar chemical environment, which also matches to the POL type structure model.

6.2 Phase transition

The examination of the phase transition was done by measuring a sequence of C1s XPS spectra while increasing the sample temperature. This sequence shows in principle that the phase transition from the α - to the β -phase occurs in a stepwise manner while heating the sample to 450 K. A close examination of the data reveals a transition in each peak from one spectrum to the next XPS spectrum. This indicates that all molecules would undergo simultaneous and analogous changes.

While this assumption may be theoretically valid, it appears to be improbable that a stepwise change occurs. A more probable theory posits that the phase occurs for some molecules prior to others, resulting in a decrease of the α -phase and an increase of the β -phase. This observation aligns with the findings reported by N. Humberg.^[16] Nevertheless, the exact process of the phase transition can not be verified with the given data, as a superposition of the two phases could not be tested.

6.3 The commensurate β -Phase

For the β -phase of QA on the Ag(100) surface one set of XPS spectra for all photoemission lines were recorded. First, the C1s XPS spectra are discussed. The β -phase exhibits four distinct peaks, that correspond to the same C atoms as in the α -phase, having the same relative positions, with the C_{arom} in the center, the C_{NH} peak towards lower, the C_{CO} - and the C_{O} peak towards higher BEs. The three peaks, C_{arom} , C_{CO} and C_{O} , have a similar shift towards lower BE upon phase transition in their BEs. This corresponds to similar changes in their chemical environment.

The lower BE could be explained with stronger adsorbate-substrate interactions. Thereby, the metal surface donates electron density to the molecule, resulting in lower BEs. This process is also observed in literature and thus a valid explanation for the observed change in the BEs.^[34]

However, the change exhibited in the C_{NH} peak is distinct. The BE remains constant during the phase transition; nevertheless, this phenomenon should not be interpreted in such a simplistic manner. Instead, it should be interpreted as a similar change in the BE towards lower values due to a stronger interaction with the surface upon phase transition and then another effect that increases the BE again. The phase

transition involves a change in the N atom bound to the C_{NH} atom, as shown in the N1s XPS spectra. The bond between the C and N atoms has a significant impact on the electron density at the C_{NH} atom and could decrease the electron density. This decreased electron density results in an increase in the BE of the C atom.

It is noteworthy, that the C atoms in the C_{NH} peak are directly next to the N atoms. For the β -phase, two different nitrogen peaks are present, which should also lead to two different C_{NH} peaks. However, the C_{NH} peak could not be resolved into two different peaks. The same applies to the C atoms next to the O atoms, as the O1s XPS spectra also exhibit two distinct peaks.

This leads to the subsequent discussion of the N1s XPS spectra for the β -phase, where the one peak for both N atoms from the α -phase divides into two different peaks. At the outset, the BE of the N1 peak undergoes a slight decrease upon the phase transition. This decrease is less significant than that observed for the other atoms. This implies that the N atom, corresponding to the N1 peak, exhibits a marginal increase in electron density, while the overall changes are analogous to those observed in the entire molecule. The observed increase in electron density may be attributed to enhanced adsorbate-substrate interactions.

The second peak for the β -phase is precisely coincident with the position of the $N1_{\text{damage}}$ peak from the α -phase, which corresponds to beam damage. It can be supposed that the N atom corresponding to the N2 peak has the equivalent chemical environment as the $N1_{\text{damage}}$ peak. As previously stated, the $N1_{\text{damage}}$ peak corresponds to a deprotonated N atom, a finding that aligns with the literature.^[45–47] This observation suggests the hypothesis that the observed N2 peak corresponds to a deprotonated N atom as well.

This means that in each QA molecule exactly one of the two amine groups is deprotonated due to the phase transition. The energy, needed for the deprotonation, has to be overcompensated by the different orientation and stronger adsorbate-substrate interaction in the β -phase, because the β -phase is thermodynamically favored.^[16] It is conceivable that the deprotonated N atoms bond to the Ag(100) surface and donate electron density to the metal, which would lead to a stabilization of the β -phase. This process could then overcompensate the energy loss due to the deprotonation and the loss of one hydrogen bond. Furthermore, the protons could interact with the Ag(100) surface and catalytically form hydrogen molecules (H_2). This process could be observable by measuring the mass of H_2 with a mass spectrometer upon the phase transition.

The transition observed from a single peak in the α -phase to two peaks in the β -phase was likewise detected in the O1s XPS spectra. The BE of the O1 peak remains nearly constant. This phenomena is analogous to that observed for the N1 peak. Consequently, it can be deduced that the N atom of the N1 peak and the O atom of

the O1 peak are the two atoms that interact with each other in the α - and β -phase. These two atoms still have a hydrogen bond in the β -phase between the keto and the amine group, which leads to similar BEs for both phases.

The second peak in the O1s XPS spectrum for the β -phase then corresponds to the other O atom that interacts with the N2 peak. This O2 peak exhibits a lower BE in comparison to the O1 peak by 0.5 eV. This corresponds to an increased electron density at the O atom, which could be result of a missing hydrogen bond between the O atom from one molecule and the hydrogen atom of an amine group from another molecule. The O atom then has a higher electron density, which leads to a lower BE.

For the β -phase, both O1s peaks in the XPS spectra have satellites. These are at 0.6 eV and 1.4 eV higher BEs than the main peaks, which corresponds to a similar BE for the O1_{sat} peak and an increase in the BE of the O2_{sat} peak compared to the α -phase. This indicates that the shake-up process also changes between the two phases.

The experimental results from the XPS spectra have to be compared to the structure model from N. Humberg.^[16] First, the structure of the β -phase is commensurate. This is also seen in the C1s XPS spectra, because it shows precise positions for each of the C atoms. Altogether, the C1s XPS spectra match to the structure model of the β -phase.

However, the O1s and N1s XPS spectra contradict the structure model. The XPS spectra show for both atom types two distinct peak with a 1 : 1 area ratio. This observation does not align with the structure model. According to N. Humberg,^[16] the structure of the β -phase has 1.5 hydrogen bonds per QA molecule, which would correspond to two distinct peaks with an area ratio of 1.5 : 0.5 (3 : 1).

Apart from this, the change in the BE for the N1s XPS spectra indicates that 1 of the two N atoms per molecule is deprotonated. This finding was not taken into account for the structure model. Furthermore, the deprotonation of one of the two N atoms would lead to the fact, that 1.5 hydrogen bonds per molecule are not possible anymore.

Consequently, the results of the experiments lead to a different structure model for the β -phase. Therefore, two theoretically possible new structure models are depicted in Figure 6.1, which match to the SPA-LEED image and STM results from N. Humberg.^[16] The structure models will be discussed in the following.

The first structure model is similar to the one of N. Humberg.^[16] The positions of the molecules are exactly the same and dimers are also formed. The most important change is the deprotonation of one of the N atoms per molecule. The most probable N atom for the deprotonation is the one that is not in between the molecules of the dimer, otherwise there would be no reason to form these dimers. In addition, the high

electron density at the deprotonated N atom and at the O atom would repel each other. This results in two hydrogen bonds between the two molecules of the dimer, meaning one hydrogen bond per molecule. No further hydrogen bonds are formed in this structure.

The structure seems the preferred one but there are further aspects that need to be discussed. In model 1, the deprotonated N atom, which then has a high electron density, is opposite to the O atom, which also has a high electron density. Usually, the negative charges should repel each other. Nevertheless, it is possible that the N and the O atom change their height and approach the surface. This could increase the instance between the two atoms. Furthermore, both atoms could donate electron density to the Ag(100) surface, which should also decrease the repulsion.

Another aspect to consider are the chemical environments of the O atoms without hydrogen bonds. There are two different chemical environments: one environment is characterized by a deprotonated amine group opposite to the O atom in the next molecule, while the other environment is opposite to the most outer ring of the QA molecule. This difference in chemical environments could lead to a changed electron density in the O atoms, which would then result in different BEs in the XPS spectra for these two environments. This was not observed in the XPS spectra. A possible explanation for this could be that the chemical environments are not sufficiently different to result in a measurable change in the BE, as the difference is only in the neighboring molecule and not in the molecule itself.

The second structure model is characterized by more changes with respect to the one of N. Humberg.^[16] Instead of the heterochiral dimers, the structure model 2 is homochiral. This means that the molecules do not have to rotate on the surface for the phase transition from the α - to the β -phase. The model also forms the same amount of hydrogen bonds as model 1.

However, one significant issue is identified with model 2. As depicted in the structure model, the O atom that is not involved in hydrogen bonding of one molecule is directly opposite to the O atom of another molecule. The plausibility of this occurrence appears to be minimal, because they should repel each other and the distance between them is so small that their van-der-Waals radii overlap. Nevertheless, it is possible that both O atoms approach the surface and donate their electron density to the surface. This could increase the distance between the O atoms and decrease the repulsion. Consequently, model 2 appears to be less probable than model 1 for the reasons previously enumerated.

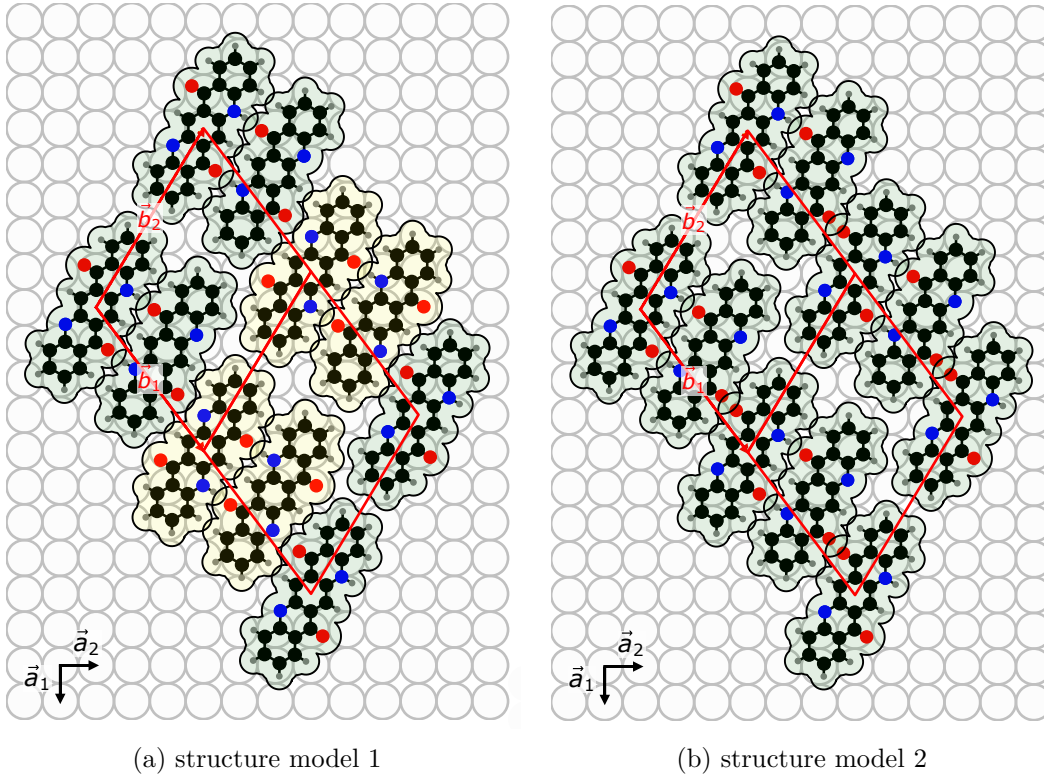


Figure 6.1: Two different structure models for the β -phase of QA on Ag(100).

7 Summary and Outlook

This study investigated the binding energies of the different atomic species and the consequential molecular orientation of QA in different phases adsorbed on the Ag(100) surface using XPS. The primary research question addressed how the QA molecules interact with the substrate and what changes in its binding situation occur upon the phase transition. Specifically, the aim was to understand the intermolecular interactions via hydrogen bonds and the adsorbate-substrate.

The XPS measurements revealed distinct peaks for the C, N and O atoms with different chemical environments within QA. The photoemission lines for C1s show different peaks for aromatic C atoms (C_{arom}), C atoms bonded amine group (C_{NH}), C atoms in carbonyl groups (C_{CO}) and C atoms bonded to oxygen (C_{O}). All these peaks were observed in both the α - and the β -phase. Furthermore, the different peaks in the C1s XPS spectra exhibit a shift towards lower BEs between the two phases, indicating an increased interaction with the substrate in the β -phase.

The N1s and O1s XPS spectra show one peak for the α -phase, whereas the β -phase exhibits two distinct peaks for the N and O atoms with an area ratio of 1 : 1. This indicates a change in the number of hydrogen bonds per molecule between the two phases, with the α -phase having two hydrogen bonds per molecule and the β -phase having one hydrogen bond per molecule. Additionally, the consequence of beam damage due to exposure of QA to the x-ray was investigated, demonstrating the deprotonation of the amine group as a result of this exposure.

As already stated, the N1s XPS spectra show a peak for the deprotonated N atoms in the β -phase, which is not present in the α -phase. This indicates that the deprotonation of one N atom per molecule occurs during the phase transition from the α - to the β -phase, explaining the change in the number of hydrogen bonds per molecule.

The results for the α -phase are consistent with the prior findings by N. Humberg.^[15,16] The proposed structure model has been confirmed and the XPS spectra for the C1s, N1s and O1s components can be subdivided into multiple peaks, which can be allocated to particular atoms of QA.

The subsequent examination of the results for the β -phase reveals inconsistencies with the proposed structure model by N. Humberg.^[15,16] It was ascertained that the β -phase is composed of two distinct O atoms, exhibiting a stoichiometric ratio of 1 : 1. A similar observation was made in the case of the N atoms. The observed shifts in the BE are indicative of the deprotonation of one of the N atoms. This results in the

formation of a single hydrogen bond per molecule. The findings of this study have led to the proposition of two novel structure model in Figure 6.1.

Nevertheless, the extant structure models leave unresolved questions. To further refine our comprehension of QA on the Ag(100) surface, it is imperative to undertake additional research to develop a more detailed structure model. Normal incidence x-ray standing wavefield absorption (NIXSW) measurements could be used to determine the adsorption height of the atoms, which allow a revision of the structure models. Beyond that, different NIXSW measurements could be used for triangulation of the β -phase. This should give exact adsorption sites for the atoms and allow a detailed structure model.

8 Appendix

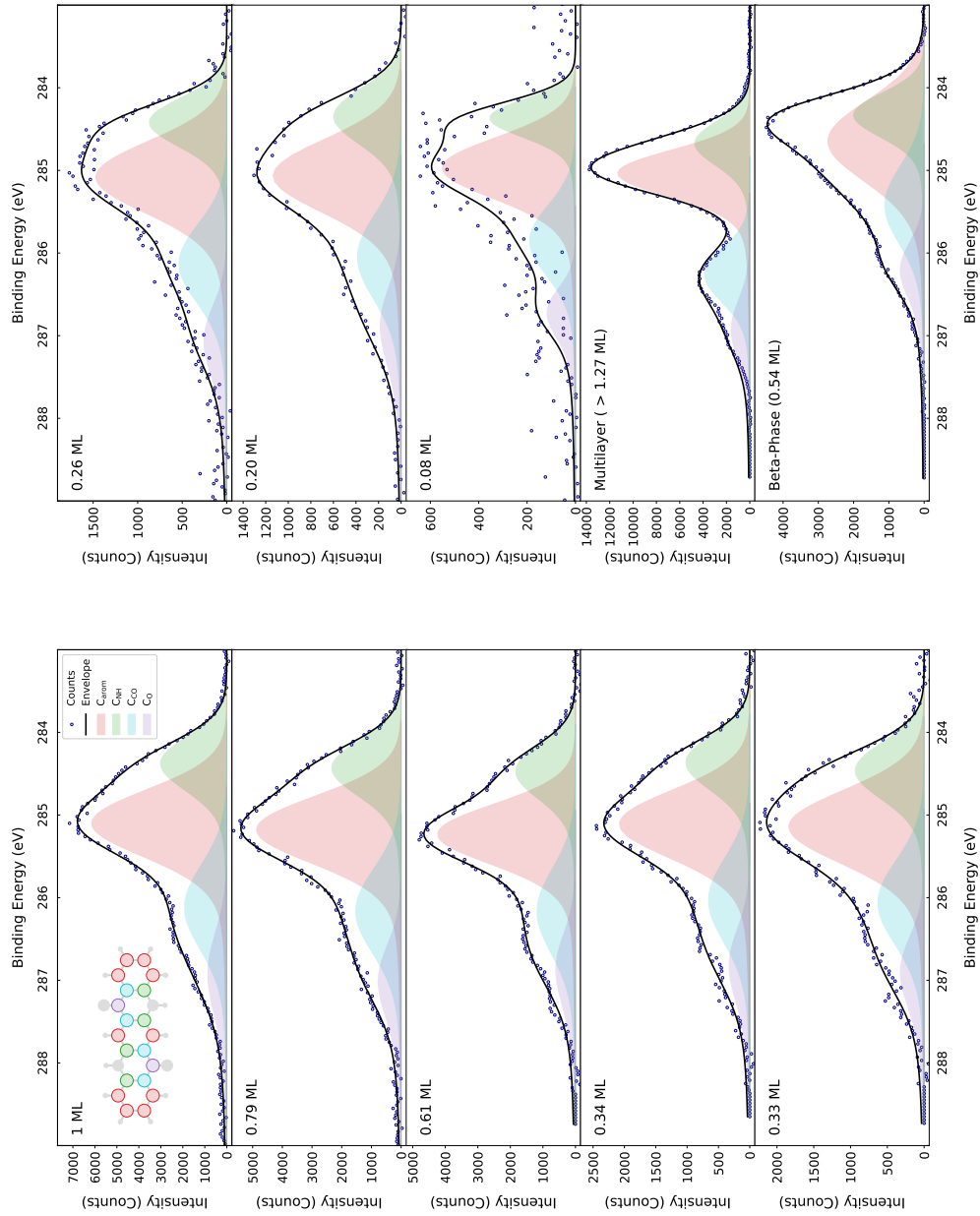


Figure 8.1: Collection of all C1s core-level XPS spectra with the different fitted peaks of all performed experiments. Each spectrum corresponds to a different preparation. The coverage of the first spectrum is set to 1 ML.

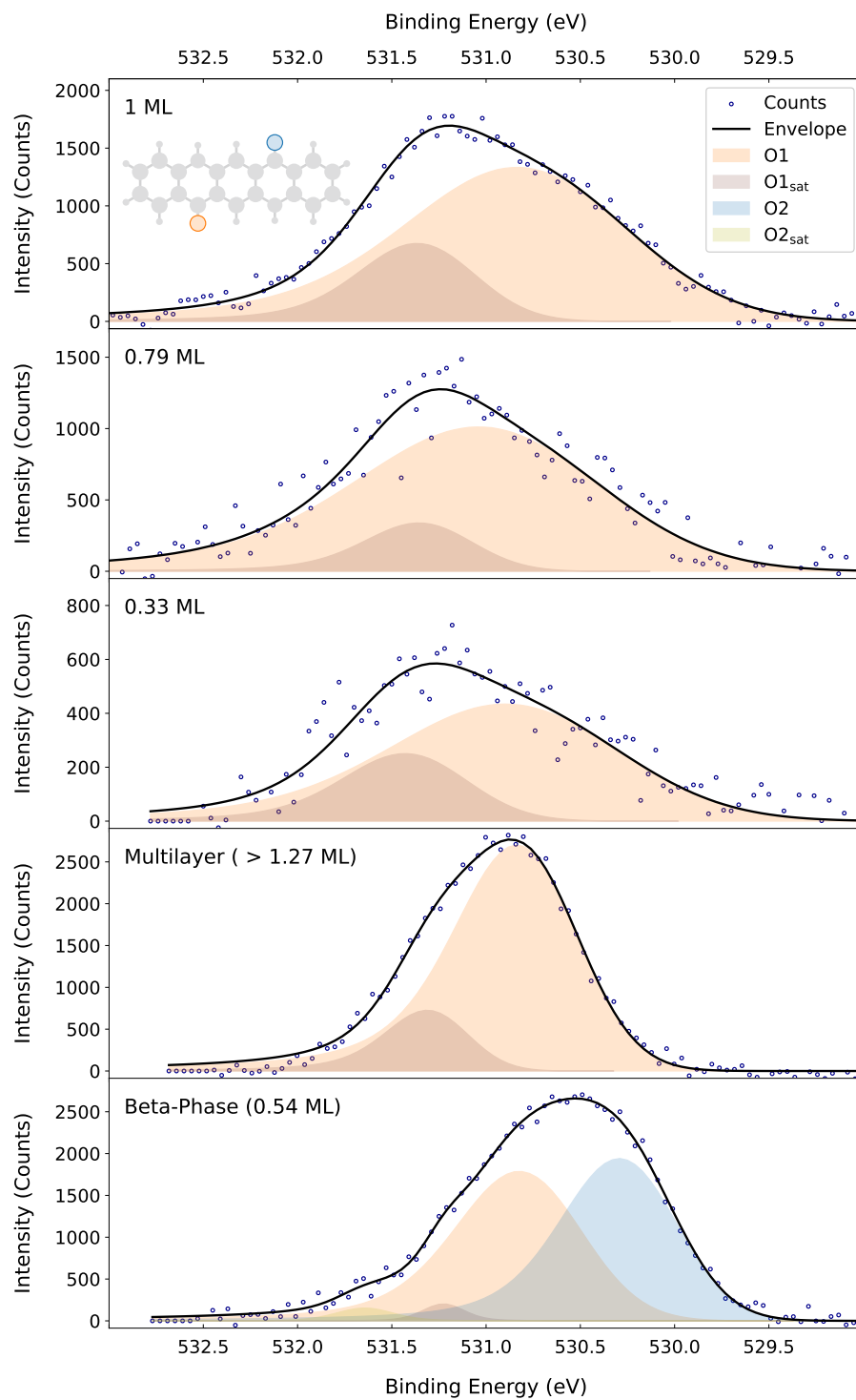


Figure 8.2: Collection of all O1s core-level XPS spectra with the different fitted peaks of all performed experiments. Each spectrum corresponds to a different preparation. The coverage of the first spectrum is set to 1 ML.

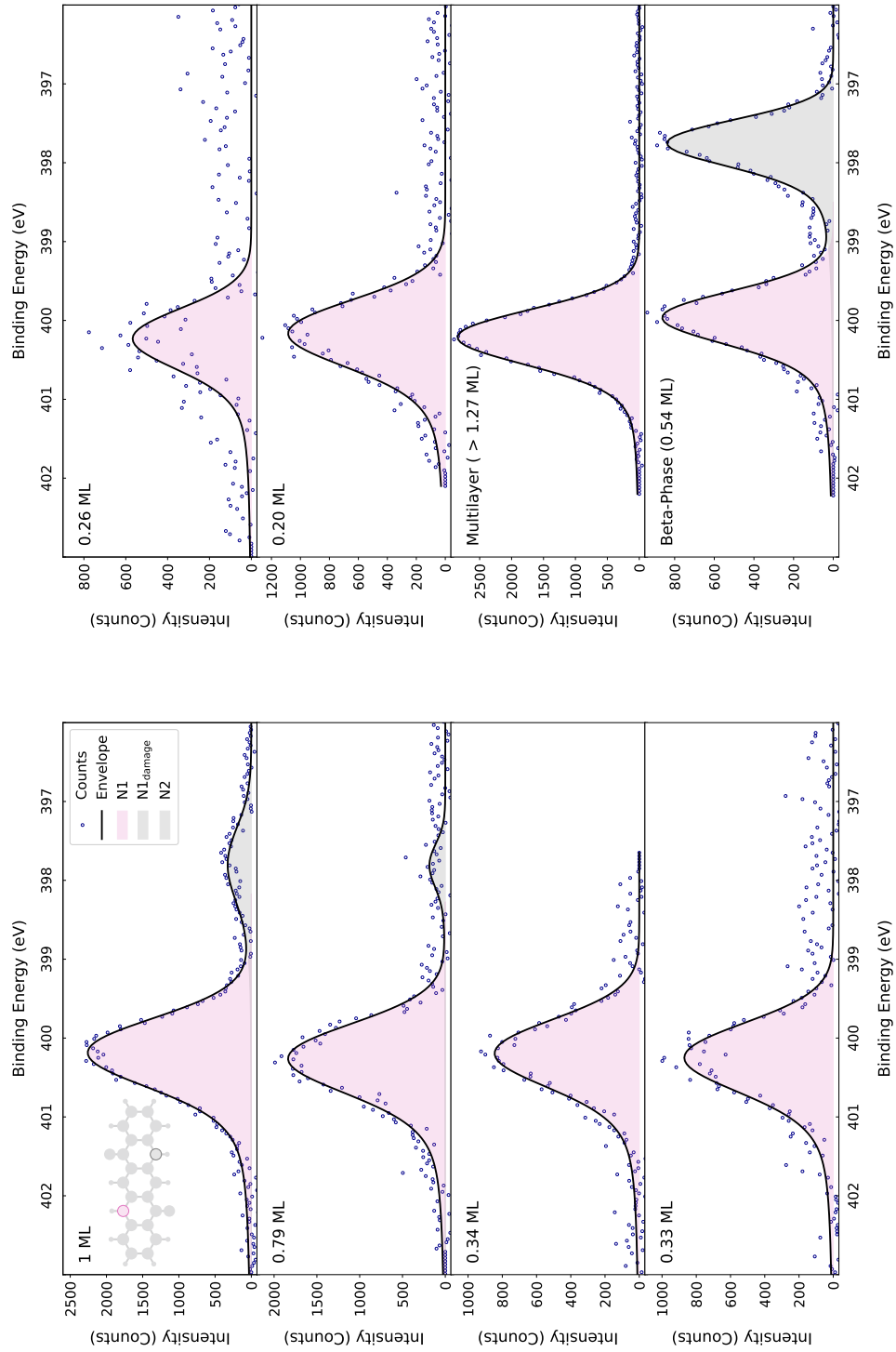


Figure 8.3: Collection of all N1s core-level XPS spectra with the different fitted peaks of all performed experiments. Each spectrum corresponds to a different preparation. The coverage of the first spectrum is set to 1 ML.

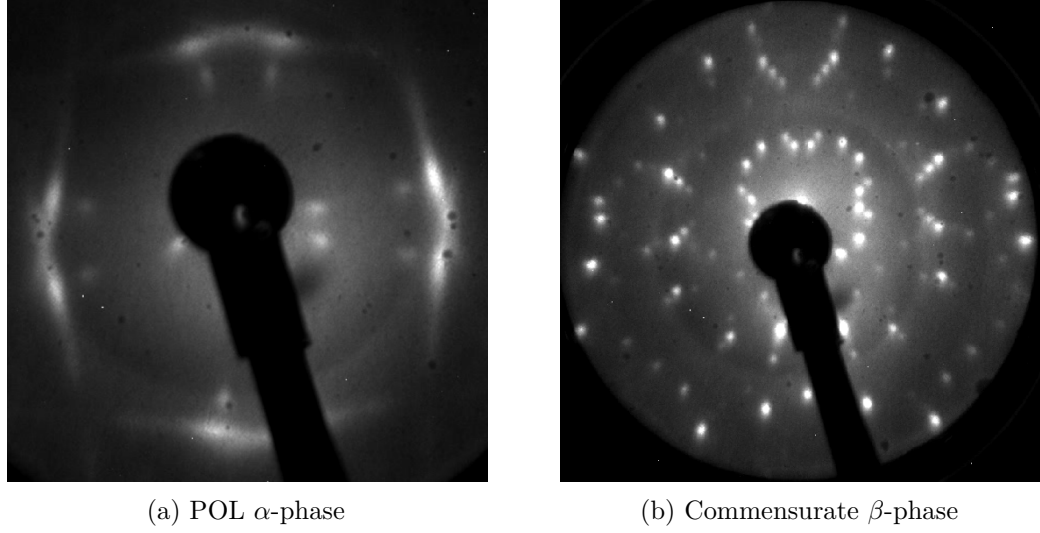


Figure 8.4: LEED patterns for the α - and the β -phase at 25.5 eV taken at the I09 beamline of the Diamond Light Source.

List of abbreviations

BE binding energy

fcc face-centered cubic

FWHM full width at half maximum

LEED low energy electron diffraction

ML monolayer

NIXSW normal incidence x-ray standing wavefield absorption

OFET organic field-effect transistor

OLED organic light-emitting diode

OPV organic photovoltaic

POL point-on-line

PTCDA perylenetetracarboxylic dianhydride

QA quinacridone

SPA-LEED spot profile analysis low energy electron diffraction

STM scanning tunneling microscopy

UHV ultra-high vacuum

XPS x-ray photoelectron spectroscopy

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